

Cryptography with Weak Privacy

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Abstract

We initiate a systematic study of information-theoretic cryptography with *weak privacy*, only requiring that the adversary cannot rule out any possible secret. For a parameter $0 < p \leq 1$, we say that a primitive has p -weak privacy (p -WP) if for every possible view V and inputs x_1, x_2 , the ratio between the probabilities of the adversary observing V on x_1 and on x_2 is at least p . This generalizes both perfect privacy, which is p -WP for $p = 1$, and a previous “combinatorial” notion of privacy, which is p -WP for *some* $p > 0$. We obtain the following main results.

- **Positive results.** We present efficient WP constructions for generalized secret sharing, decomposable randomized encodings, and the related notions of garbling schemes and PSM protocols, as well as interactive secure multiparty computation protocols in the plain model and in the OT-hybrid model.

For secret sharing, we settle a question of Beimel and Franklin (TCC 2007), showing that every n -party access structure admits a WP scheme with per-party share size $O(n)$. When all unauthorized sets have constant size, we get a p -WP scheme with constant share size and $p \geq 1/\text{poly}(n)$.

- **Negative result.** For decomposable randomized encodings, we show that a previous lower bound technique of Ball et al. (ITCS 2020) applies also to the WP notion. Together with our upper bound, this shows that the optimal WP garbling size of the worst-case $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is $\Theta(n^2)$.
- **Application.** While WP may seem like an unrealistically weak security notion, we demonstrate its usefulness towards achieving traditional security guarantees. Concretely, under the standard LPN assumption, we show that any p -WP secret-sharing scheme with inverse-polynomial p implies a *computationally secure* secret-sharing scheme for a related access structure. Together with our positive results for WP secret sharing, this implies a super-polynomial improvement of the share size for a natural class of access structures.

1 Introduction

Information-theoretic cryptography is an area full of big open questions. For the widely studied problem of information-theoretic secure multiparty computation (MPC), there are still exponential gaps between the best upper and lower bounds on the communication required for computing general functions $f : \{0, 1\}^n \rightarrow \{0, 1\}$ [10]. Similarly, for the problem of realizing a general monotone access structure Γ by a secret-sharing

scheme, there are still exponential gaps between the best known lower bound on the share size [22] and the best known upper bounds, in spite of significant recent progress on the upper bounds front [39, 40, 38, 2, 4].

Motivated by this state of affairs, it is natural to consider simpler “toy versions” of the above open questions. In this work, we revisit these questions by considering a very weak notion of security that only insists on *possible deniability*, in the sense that no admissible secret can be ruled out by the adversary. Can the above exponential gaps be closed in this simpler setting?

Weak privacy. More concretely, we consider the following notion of weak secrecy which is inspired by differential privacy. For $0 < p \leq 1$, we say that a primitive is *p-weakly private*, or *p-WP* for short, if for any possible view V of the adversary and any two “equivalent” inputs x_1, x_2 ,

$$\Pr[\text{The adversary sees } V \text{ on input } x_1] \geq p \cdot \Pr[\text{The adversary sees } V \text{ on input } x_2].$$

For example, in the context of secret-sharing schemes, the view of an adversary consists of the shares of an unauthorized set of parties, and the inputs are any two secrets. This is similar to the definition of pure differential privacy of Dwork, McSherry, Nissim, and Smith [25] except for considering all pairs of admissible inputs rather than only pairs of neighboring inputs. However, in the differential privacy literature, the parameter p is typically viewed as a constant.¹ In contrast, we typically consider small values $p > 0$ that vanish with the instance size n (e.g., $1/n$ or even $1/2^n$). While potentially leaking a lot of information about the inputs, WP implementations of cryptographic primitives still leave some small ambiguity and hence a possible room for deniability.

Jumping ahead, we will later use the fact that a p -WP view of a secret bit b can be simulated from the output of a *binary symmetric channel* that flips b with probability $\delta = p/(1 + p)$ [33], which is close to p when p is small. This motivates the specific distance metric that we use. In contrast, the standard notion of security with statistical distance $0 \leq \varepsilon \leq 1$ can only guarantee a similar simulation from the output of a *binary erasure channel* that erases b with probability $\delta = 1 - \varepsilon$ [41].

Example 1.1 (WP private information retrieval). In a 2-server private information retrieval (PIR) protocol [20], two servers hold an n -bit database and a user wants to retrieve the i -th bit of the database without disclosing i to each server. The best known 2-server PIR protocol with perfect privacy has communication $2\tilde{O}(\sqrt{\log n})$ [24]. We next describe a simple WP 2-server PIR protocol with much better efficiency; the user, on index i , with probability $1/2$ sends i to the first server and a random index $j \neq i$ to the second server and with probability $1/2$ sends i to the second server and a random index $j \neq i$ to the first server. Each server sends the bit of the database specified by the message it gets. The probability that a server gets the index j when the user’s index is j is $1/2$ and the probability that a server gets the index j when the user’s index is $i \neq j$ is $1/2(n - 1)$; thus, the protocol is $\frac{1}{n-1}$ -WP.

In the extreme, we consider a “combinatorial” notion of secrecy, sometimes referred to as *weak privacy* (WP), which only requires that it is *never* the case that an adversary can rule out an input with *absolute certainty*. Equivalently, WP is p -WP for an arbitrary $p > 0$. Such a combinatorial notion of secrecy was already studied for different cryptographic problems. Beimel and Franklin [9] studied weakly private secret-sharing schemes, requiring that any set of shares which is possible for an unauthorized set of parties on one secret s is also possible on any other secret s' (i.e., the two share distributions have equal supports, though the probabilities can be very different). Beaver, Haber, and Winkler [8] considered key agreement from correlated randomness, where they only require weak privacy, i.e., every key is *possible* given the communication that is exchanged between the parties.

¹Since here we consider a minimal notion of privacy, we use a different parameterization than in the differential privacy literature; namely, p -WP corresponds to ε -DP for $\varepsilon = \ln(1/p)$.

Motivation. Beyond the theoretical interest in WP as a natural notion of security, we study this notion for three additional reasons. The first two also apply to the “combinatorial” notion of weak privacy, while the third critically depends on the quantitative notion of p -WP we introduce in this work.

- **STRONGER NEGATIVE RESULTS.** Since weak privacy relaxes perfect privacy, ruling out weak privacy yields stronger negative results that apply to the *combinatorics* of the problem at hand. Beimel and Franklin [9] give several examples of cryptographic tasks where weak privacy and perfect privacy have essentially the same power; this gives a more complete picture of the feasibility of such tasks. For example, Shannon’s classic result [43] that in any perfect encryption scheme the key length is at least the length of the message only requires weak privacy (i.e., that given an encryption, all messages are possible). As another example, Kushilevitz [37] characterized the functions that can be computed with perfect privacy by two semi-honest parties; in particular, a function can be computed in this model with perfect privacy if and only if it can be computed with weak privacy.
- **GUIDING RESEARCH EFFORTS.** When considering challenging open questions about perfect privacy, studying the simpler combinatorial notion of weak privacy is a natural first step. In particular, WP lower bounds (while stronger) are typically simple to obtain whenever they exist. In light of this, strong *upper bounds* for weak privacy can be useful in ruling out such a combinatorial approach to lower bounds. For example, for the case of the 2-out-of- n threshold access structure, Beimel and Franklin showed that 2 bits per share suffice for a weakly-private secret sharing of a 1-bit secret, compared to the $\Theta(\log n)$ share size required for a (standard) perfectly private scheme for this access structure [42, 15, 34, 18]. Alternatively, if improved WP upper bounds turn out to be challenging, this may suggest that the barrier towards progress is combinatorial in nature.
- **BOOSTING WP TO STANDARD SECURITY.** Somewhat surprisingly, we will demonstrate that in some cases p -WP security with an inverse-polynomial p can serve as a useful step towards new results for standard cryptographic primitives satisfying the usual notion of security.

1.1 Our Contribution

Motivated by the above, we initiate a systematic study of WP security for some of the central primitives in information-theoretic cryptography. In a nutshell, we obtain three types of results:

- **POSITIVE RESULTS.** We present efficient WP constructions for secret-sharing schemes realizing general access structures, decomposable randomized encodings (and the related notions of garbling schemes and PSM protocols), and secure two-party and multi-party computation with and without correlated randomness.
- **NEGATIVE RESULT.** For the case of decomposable randomized encodings, we show that a previous lower bound technique of Ball, Holmgren, Ishai, Liu, and Malkin [6] applies also to the relaxed WP notion. This leads to an almost tight bound on the WP complexity of this and related primitives.
- **APPLICATION.** We show how to leverage our positive results on WP secret sharing for general access structures to improve the state of the art on *computational* secret sharing for a natural class of access structures (under a standard cryptographic assumption).

We now give a more detailed account of our results, starting with the case of secret sharing.

WP secret-sharing schemes. In a secret-sharing scheme [42, 15, 32], there is a dealer holding a secret (for simplicity, a bit) that she wants to distribute among a set of parties. This is done by giving each party a string (called the share) such that only some pre-defined subsets of the n parties can recover the secret from their shares; the collection of authorized sets is called an access structure. In a perfect secret-sharing scheme, the probability of any possible shares of an unauthorized set (a set not in the access structure) given that the secret is 0 is equal to the probability that these shares are generated for the secret 1. Our WP definition allows for the probabilities of the shares for the two secrets 0, 1 to be different, but it bounds the ratio between them. Formally, for $0 < p \leq 1$, the notion of p -WP requires that for every unauthorized set A , any possible collection of shares $\langle sh_j \rangle_{p_i \in A}$, and any secret $s \in \{0, 1\}$ we have:

$$\begin{aligned} & \Pr[\text{The shares generated for } A \text{ on the secret } s \text{ are } \langle sh_j \rangle_{p_i \in A}] \\ & \geq p \cdot \Pr[\text{The shares generated for } A \text{ on the secret } \bar{s} \text{ are } \langle sh_j \rangle_{p_i \in A}]. \end{aligned}$$

In particular, $p = 1$ gives perfect privacy, while any positive $p > 0$ gives weak privacy. We present our results for secret-sharing schemes in Section 3. The main result is a p -WP secret-sharing scheme with linear share size, for every access structure. When all unauthorized sets are of constant size, we get p which is inverse-polynomial; otherwise, it can be exponentially small. More generally:

Informal Theorem 1 (WP secret sharing for general access structures). *Let Γ be an n -party access structure in which all unauthorized sets are of size at most t . Then, there is a p -WP secret-sharing scheme (for 1-bit secrets) that realizes Γ with per-party share size $O(t)$, where $p = \Omega(n^{-t})$ if t is constant and $p = 2^{-O(t \log(n/t))}$ for general $t = t(n)$.*

Access structures where the size of the maximal unauthorized sets is $t = 2$ are known as forbidden graph access structures [44]. The best known perfect secret-sharing scheme for n -party forbidden graph access structures has shares of size $2^{\tilde{O}(\sqrt{\log n})}$ [39]. We construct a $1/n^2$ -WP scheme for them with share size $O(1)$. For general n -party access structures, the best known perfect secret-sharing scheme has shares of size $2^{0.58n}$ [4]. We construct a $1/2^{\Theta(n)}$ -WP scheme for them with share size $O(n)$. This solves an open problem of Beimel and Franklin [9].

We shortly recall previous results on weakly-private secret-sharing schemes. The work of Beimel and Franklin [9], which served as the starting point for our work, defined and studied weakly-private secret-sharing schemes. When combined with [4], their construction yield, for every access structure, a WP secret-sharing scheme for sharing a long q -bit secrets with share size $q + 2^{0.58n}$; that is, their scheme has information ratio 1 for large values of q . In contrast, we construct such schemes with share size $q + O(n)$. Beimel and Franklin [9] also constructed WP t -out-of- n secret-sharing schemes with share size $O(t)$, which improves upon perfect secret-sharing schemes when $t < \log n$. In addition, Beimel and Franklin proved that the share size in any weakly-private t -out-of- n secret-sharing scheme is at least $\Omega(\min\{t, \log \log \log(n - t)\})$. This implies that the share size in weakly-private secret-sharing schemes for the worst access structure is $\Omega(\log \log \log(n)) = \omega(1)$. A related notion of differentially-private secret-sharing schemes was previously studied by Talwar [45], where a differentially-private n -out-of- n secret-sharing scheme of vectors of norm at most 1 was constructed.

WP decomposable randomized encoding. We then turn to apply this notion of weak privacy in the context of Decomposable Randomized Encoding (DRE) (see [3]). A DRE can be viewed as a simple information-theoretic variant of a garbling scheme [48, 13], or alternatively as an n -party Private Simultaneous Message (PSM) protocol [27, 31] in which each party holds only one input bit.

More formally, a randomized encoding of a function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is a randomized function ENC that, given an input $x \in \{0, 1\}^n$, outputs an encoding $\hat{y} \leftarrow \text{ENC}(x; r)$ of $f(x)$ while hiding all other information about x . That is, (1) one can recover $f(x)$ from the encoding $\hat{y}$, i.e., there is a decoding procedure DEC such that $\text{DEC}(\text{ENC}(x; r)) = f(x)$, and (2) $\text{ENC}(x; r)$ and $\text{ENC}(x'; r)$ are equally distributed for every x, x' such that $f(x) = f(x')$. Of course, $\text{ENC}(x)$ can always output $f(x)$; however, usually the goal is to use a randomized function ENC that is easier to compute than the original function f . We consider a scenario in which $x = (x_1, \dots, x_n)$ is distributed between n parties, i.e., the i -th party holds x_i and there is no central authority that can compute $\text{ENC}(x)$. In a *decomposable randomized encoding (DRE)*, each party locally computes $\text{ENC}_i(x_i; r)$ such that $\text{ENC}(x; r) = (\text{ENC}_1(x_1; r), \dots, \text{ENC}_n(x_n; r))$ is a randomized encoding of x (note that all local encodings have access to the same randomness r). The best known DRE for an arbitrary function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is of size $\text{poly}(n) \cdot 2^{n/2}$ [10, 11, 5]. We consider p -WP decomposable randomized encodings, where the privacy requirement is relaxed, requiring that $\Pr[\text{ENC}(x; r) = \hat{y}] \geq p \cdot \Pr[\text{ENC}(x'; r) = \hat{y}]$, for every x, x' such that $f(x) = f(x')$ and every $\hat{y}$. We show how to construct efficient WP randomized encoding for arbitrary functions.

Informal Theorem 2 (Upper bound for WP DRE). *Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be a function. There is a $2^{-O(n)}$ -WP decomposable randomized encoding for f with encoding size $\tilde{O}(n^2)$.*

We also prove nearly matching lower bounds for WP randomized encoding of some explicit functions, showing that the encoding size in the above theorem is tight up to a logarithmic factor. This lower bound is the same as the lower bound of Ball et al. [6], which is the best known lower bound for statistical randomized encoding; our proof is an adaptation of the proof of Ball et al., showing that weak privacy suffices to prove the lower bound. We note that our result is incomparable to the results of Ball et al., as their proof applies to the incomparable notion of statistical privacy and allows error in the correctness.

Informal Theorem 3 (Lower bound for WP DRE). *There exist explicit functions $f : \{0, 1\}^n \rightarrow \{0, 1\}$ such that, for every $p > 0$, any p -WP decomposable randomized encoding for f has size $\Omega(n^2 / \log n)$.*

WP secure MPC protocols. We construct WP k -party secure multiparty computation protocols, for any function $f : (\{0, 1\}^n)^k \rightarrow \{0, 1\}$, that are secure against a semi-honest adversary. We construct such protocols in two settings. First, we consider the plain model and assume that there is an honest majority, i.e., the adversary controls less than $k/2$ parties. Second, we consider the case where there is no guarantee of an honest majority, i.e., the adversary can control any subset of parties. In this case, for many functions, MPC protocols do not exist in the plain model even if we require weak privacy; the impossibility results of [21, 37] hold also assuming weak privacy. We, therefore, consider the OT-hybrid model, i.e., in each step of the protocol a pair of parties can give inputs to an ideal computation of the oblivious transfer (OT) functionality and the first party gets its output.² The p -WP property for these two models is defined analogously to the case of DRE described above. We obtain the following results.

Informal Theorem 4 (Upper bound for WP MPC protocols). *Let $f : (\{0, 1\}^n)^k \rightarrow \{0, 1\}$ be a function.*

- *There is a $2^{-O(kn)}$ -WP secure k -party protocol for f in the plain model assuming an honest majority, in which the communication complexity is $\text{poly}(k) \cdot n$.*
- *There is a $2^{-O(kn)}$ -WP secure k -party protocol for f in the OT-hybrid model secure against any number of semi-honest parties, in which the communication complexity and the number of calls to OT are $\text{poly}(k) \cdot n$.*

²In an ideal computation of OT, the first party's input is $i \in \{0, 1\}$, the second party's input is a pair of bits b_0, b_1 , the first party gets output b_i without learning any information about $b_{\bar{i}}$ and the second party does not learn any information about i .

Beimel and Malkin [12] proved that in any secure 2-party protocol for the equality function with inputs of size n (and for other functions) the OT complexity is $\Omega(n)$; this lower bound holds also for weakly-private 2-PC protocols (see Appendix A) and implies a lower bound of $\Omega(kn)$ for some functions $f : (\{0, 1\}^n)^k \rightarrow \{0, 1\}$ (e.g., $f(x_1, \dots, x_k) = 1$ if $x_1 \circ \dots \circ x_{k/2} = x_{k/2+1} \circ \dots \circ x_k$ for an even k). This implies that our weakly-private protocols in the OT-hybrid model are optimal up to a factor of $\text{poly}(k)$.

We next review the best known protocols with perfect privacy for this task, starting with the OT-hybrid model and then moving to the plain model with honest majority. Beimel et al. [10] showed that every function $f : (\{0, 1\}^n)^2 \rightarrow \{0, 1\}$ can be computed by a secure 2-PC protocol in the OT-hybrid model, where the OT and communication complexities are $2^{\tilde{O}(\sqrt{n})}$ (this is achieved by plugging the PIR protocol of [24] in the protocol of [10]). For $k > 2$, the best k -party MPC protocol in the OT-hybrid model for a function $f : (\{0, 1\}^n)^k \rightarrow \{0, 1\}$ is the GMW protocol [30], where the OT and communication complexities are $\text{poly } k$ times the size of the best Boolean circuit computing f ; for almost all functions this results in a protocol with OT and communication complexities $2^{O(kn)}$.

The best known secure 3-PC protocol for a function $f : (\{0, 1\}^n)^3 \rightarrow \{0, 1\}$ in the plain model assuming at most one semi-honest party has communication complexities $2^{\tilde{O}(\sqrt[3]{n})}$ (this is achieved by plugging the PIR protocol of [29] in the 2-PC protocol of [10] with correlated randomness and transforming it into a secure 3-party protocol in the plain model). For $k > 3$, the best k -party MPC protocol for a function $f : (\{0, 1\}^n)^k \rightarrow \{0, 1\}$ in the plain model is the BGW protocol [14], where the communication is $\text{poly } k$ times the size of the best Arithmetic circuit computing f ; for almost all functions this results in a protocol with communication complexity $2^{O(kn)}$.

Computational secret-sharing schemes for composed gap access structures. Finally, we present a somewhat unexpected application of our new WP secret-sharing schemes. While it may seem that WP security is unrealistically weak, we show that this notion (with inverse-polynomial p) can be leveraged towards improving the efficiency of *standard* computational secret sharing for a natural class of access structures. In fact, we obtain an exponential improvement over the best known information-theoretic secret-sharing schemes for the same access structures. Recall that in computational secret-sharing schemes [49, 36, 1], the sharing and reconstruction are efficient, and a computationally-bounded adversary cannot learn information from the shares of an unauthorized set.

We consider “composed gap access structures.” To explain what this means, recall the notion of ramp secret-sharing schemes [16], which generalize threshold schemes. Ramp schemes (also known as gap threshold schemes) are parameterized by the number of parties n and two integers $0 \leq \alpha \leq \beta \leq 1$; it is required that every set of less than αn parties should not learn any information about the secret, while every set of size at least βn parties can reconstruct the secret. This scheme realizes a partially-defined access structure, i.e., there are no requirements on sets of parties whose size is at least αn and less than βn . We consider a composed gap access structure, where we replace each party in the ramp scheme by an access structure; that is, we are given m access structures $\Gamma_1, \dots, \Gamma_m$ on disjoint sets of parties $P_1, \dots, P_m$ and parameters $0 \leq \alpha \leq \beta \leq 1$ and consider a composed gap access structure $\text{Gap}_{\alpha, \beta}(\Gamma_1, \dots, \Gamma_m)$, which is a partially defined access structure such that:

- A set B is authorized if it is authorized in at least βm access structures, i.e., $|\{i : B \cap P_i \in \Gamma_i\}| \geq \beta m$.
- A set A is unauthorized if it is authorized in less than αm access structures, i.e., $|\{i : A \cap P_i \in \Gamma_i\}| < \alpha m$.

As in ramp schemes, we do not have any requirements on sets C such that $\alpha m \leq |\{i : C \cap P_i \in \Gamma_i\}| < \beta m$ (i.e., they can either reconstruct the secret, learn no information about the secret, or learn partial information

about the secret). Composed-gap access structures are used in [2, 4] to construct the best known schemes for arbitrary access structures; in their construction, each Γ_i has exponentially many parties, while we only construct schemes where each Γ_i has a logarithmic number of parties.

Our computational secret-sharing schemes are based on the hardness of the learning parity with noise (LPN) problem [17]. The (decisional) LPN assumption roughly says that, for a public random linear code generated by a random matrix $C \in \mathbb{F}_2^{O(k) \times k}$, it is hard to distinguish between a noisy encoding $Cx + e$ of a random message $x \in \mathbb{F}_2^k$ and a uniformly random vector in $\mathbb{F}_2^{O(k)}$. Concretely, we assume that each coordinate of the error vector e is independently sampled from a Bernoulli distribution with parameter $1/\sqrt{k}$. In fact, LPN is widely conjectured to hold even if we take a Bernoulli distribution with parameter $1/k^c$ for any constant $0 < c < 1$.

Informal Theorem 5 (Computational secret-sharing schemes for composed gap access structures). *Let α, β be constants such that $\beta > 1/2$ and $0 \leq \alpha \leq 1 - h(\beta)$. There exists a constant γ such that assuming that $\text{LPN}(k, \gamma/\sqrt{k})$ holds, the following computational secret-sharing schemes exist:*

- A scheme with share size $O(1)$ for the family of access structures $\text{Gap}_{\alpha, \beta}(\Gamma_1, \dots, \Gamma_m)$, where each Γ_i is a forbidden graph access structure with $m^{1/4}$ parties.
- A scheme with share size $\log m$ for the family of access structures $\text{Gap}_{\alpha, \beta}(\Gamma_1, \dots, \Gamma_m)$, where each Γ_i is an access structure with $0.1 \cdot \log m$ parties.

We next compare the above computational schemes to the best known perfect secret-sharing schemes for composed gap access structures. For $\beta = 1$ and a constant $\alpha < 1$, there is a perfect secret-sharing scheme for composed gap access structures for forbidden graph access structures with 1-bit shares (see Section 4.1). For other cases, the known perfect schemes are much less efficient than the computational schemes; for composed gap forbidden graph access structures, the best scheme we can construct has share size $2^{\tilde{O}(\sqrt{\log n})}$ and for the gap composition of arbitrary access structures, the best scheme we can construct has share size $n^{0.058-o(1)}$. That is, for the latter case, there is an exponential gap in the share size of our WP secret-sharing scheme and the best known perfect secret-sharing scheme.

Open Problems. We initiated a systematic study of WP variants of central cryptographic primitives. In addition to extending this study to other primitives, our results leave several natural open problems.

- What is the biggest $p \leq 1$ such that there are p -WP secret-sharing schemes for arbitrary access structures with polynomial share size? As the best known lower bound on the share size for perfect secret-sharing schemes is $\Omega(n/\log(n))$ [22], we cannot rule out that perfect secret-sharing schemes (i.e., $p = 1$) with polynomial share size exist for every access structure.
- Are there WP secret-sharing schemes for arbitrary access structures with share size $o(n)$?
- Graph access structures are access structures in which the minimal authorized sets are of size two (however, unauthorized sets can be big). The best upper bound on the share size for graph access structures is $O(n/\log n)$ [26], which is better than our construction applied to graph access structures, when there are large independent sets in the graph. Are there more efficient WP secret-sharing schemes for graph access structures?
- Beimel and Franklin [9] constructed $1/n$ -WP 2-out-of- n threshold secret-sharing schemes with share size 2, leaving the case of large thresholds open. Is there a WP $(n-1)$ -out-of- n threshold secret-sharing scheme with share size $o(\log n)$?

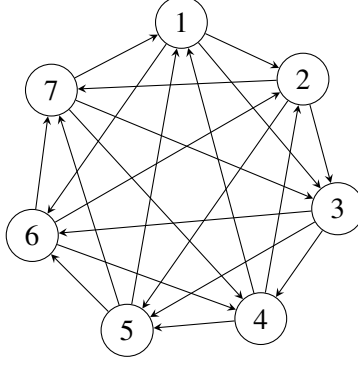


Figure 1: A 2-dominated graph with 7 vertices.

- Finally, it would be interesting to find additional applications of WP security in mainstream cryptography.

1.2 Technical Overview

1.2.1 Weakly Private Secret Sharing for Arbitrary Access Structures

Warm-up. A forbidden graph access structure [44] is an access structure in which all maximal unauthorized sets are of size 2. We start with a simple $\Theta(1/n^2)$ -WP secret-sharing scheme realizing any forbidden graph access structure with 1-bit shares but with imperfect reconstruction. In this scheme, an authorized set either reconstructs the correct value or it outputs $\perp$ (it never outputs an incorrect value for the secret). The scheme is simple: to share a secret $s \in \{0, 1\}$, the dealer chooses with uniform distribution a maximal unauthorized set $\{p_{i_1}, p_{i_2}\}$ (of size 2) and two bits r_1, r_2 , gives the secret s as the share to all parties not in $\{p_{i_1}, p_{i_2}\}$, gives r_1 to party p_{i_1} , and gives r_2 to p_{i_2} . Reconstruction by an authorized set $B \in \Gamma$ is as follows: if the shares of all parties in B are the same value b , then the parties in B output b as the secret; otherwise, they output $\perp$. Since Γ is monotone, $B \setminus \{p_{i_1}, p_{i_2}\} \neq \emptyset$ and at least one party in B gets the secret as its share and so the reconstruction never errs.³ For the WP property of the scheme, consider an unauthorized set $A \notin \Gamma$. With probability greater than $1/n^2$, this set is chosen by the dealer and the shares of the two parties in the set are uniformly distributed in $\{0, 1\}^2$. Thus, the probability of each vector of shares in $\{0, 1\}^2$ is at least $O(1/n^2)$; in particular, the scheme is $O(1/n^2)$ -WP.

A $1/O(n^2)$ -WP secret-sharing scheme for forbidden graphs. In the scheme described above, when a minimal authorized set B gets two different shares, it does not know which one of them is the correct secret. To get perfect correctness, still with constant size shares, we add a mechanism that enables any authorized set to identify a party holding the correct value. This is done using a tournament $G_0 = (V_0, E_0)$ ⁴ such that for every $v_1, v_2 \in V_0$ there is a vertex $u \neq v_1, v_2$ such that $(u, v_1), (u, v_2) \in E_0$, i.e., u “dominates” v_1, v_2 . Such a tournament, called a 2-dominated graph, is described in Figure 1; it has 7 vertices. For each pair of vertices, there is a unique vertex that dominates them. For example, $(1, 2), (1, 6) \in E_0$; thus, vertex 1 dominates vertices 2, 6.

³The probability that an authorized set B outputs $\perp$ can be reduced to $O(1/n)$; see details in Section 3.2.

⁴A directed graph $G_0 = (V_0, E_0)$ is a tournament if for every distinct $u, v \in V_0$ either $(u, v) \in E_0$ or $(v, u) \in E_0$ (but not both).

We use a 2-dominated graph $G_0 = (V_0, E_0)$ to construct a perfectly correct WP secret-sharing scheme for a forbidden graph access structure determined by an n -vertex graph $G = (V, E)$ as follows:

- On secret $s \in \{0, 1\}$, the dealer chooses at random, with uniform distribution, a maximal unauthorized set $A = \{p_i, p_j\}$, uniformly distributed vertices $v_i, v_j \in V_0$, and uniformly distributed secrets $s_i, s_j \in \{0, 1\}$.
- Let $u \neq v_i, v_j$ be a vertex in G_0 that dominates v_1, v_2 ; for every $p_\ell \notin A$ let $s_\ell = s$ and $v_\ell = u$.
- The share of p_ℓ , for $\ell \in [n]$, is $\langle s_\ell, v_\ell \rangle$.
- Reconstruction for a minimal authorized set $B = \{p_{i_1}, p_{i_2}\} \in \Gamma$ holding shares $\langle s_{i_1}, v_{i_1} \rangle$ and $\langle s_{i_2}, v_{i_2} \rangle$: If $v_{i_1} = v_{i_2}$ or $(v_{i_1}, v_{i_2}) \in E_0$ output s_{i_1} , otherwise output s_{i_2} .

We first explain why the reconstruction is always correct. By the monotonicity of Γ , we have $B \setminus A \neq \emptyset$. If $p_{i_1}, p_{i_2} \notin A$, then they both hold the share $\langle s, u \rangle$ and the reconstruction outputs the correct secret s . If $p_{i_1} \notin A$ and $p_{i_2} \in A$, then party p_{i_1} holds the share $\langle s, u \rangle$, party p_{i_2} holds the share $\langle s_{i_2}, v_{i_2} \rangle$, and $(u, v_{i_2}) \in E_0$, and the reconstruction outputs the correct secret $s = s_{i_1}$. If $p_{i_1} \in A$ and $p_{i_2} \notin A$, then party p_{i_1} holds the share $\langle s_{i_1}, v_{i_1} \rangle$, party p_{i_2} holds the share $\langle s, u \rangle$, and $(v_{i_1}, u) \notin E_0$ (since G_0 is a tournament and $(u, v_{i_1}) \in E_0$); again, the reconstruction outputs the correct secret $s = s_{i_2}$.

We claim that this scheme is $1/O(n^2)$ -WP. We first argue that the scheme is weakly private. Let $A' = \{p_{i'}, p_{j'}\}$ be a maximal unauthorized set. Every pair of shares $\langle s_{i'}, v_{i'} \rangle, \langle s_{j'}, v_{j'} \rangle \in (\{0, 1\} \times V_0)^2$ is possible, for every secret s , when the dealer chooses $A = A'$ as the random unauthorized set, chooses these shares as the shares of A' , takes a vertex u such that $(u, v_{i'}), (u, v_{j'}) \in E_0$ (such u exists by the properties of G_0), and gives the share $\langle s, u \rangle$ to all the other parties. Since the probability of each vector of shares is at least $\frac{1}{\binom{n}{2} \cdot (|V_0| \cdot 2)^2} = \frac{1}{O(n^2)}$, the scheme is $1/O(n^2)$ -WP.

A WP secret-sharing scheme for arbitrary access structures with share size $O(n)$. Our WP secret-sharing scheme for arbitrary access structures Γ is a generalization of the scheme for forbidden graphs. Let $t < n$ be the maximum size of an unauthorized set. We use a tournament $G_0 = (V_0, E_0)$ such that for every $W \subset V_0$ of size t there is a vertex $u \in V_0$ that dominates all vertices in W , i.e., $(u, v) \in E_0$ for every $v \in W$. We prove that a random tournament with $O(t^2 2^t)$ vertices has this property with high probability. The share of each party contains a secret and a vertex of the tournament, i.e., the share length is $O(t) = O(n)$. A simple implementation of this idea will result in a $2^{-O(n^2)}$ -WP secret-sharing scheme. We describe a better distribution on the vertices and secrets given to the parties in A that results in a $2^{-O(n)}$ -WP secret-sharing scheme in Section 3.

1.2.2 LPN and Computational Secret Sharing Schemes for Composed Access Structures

We construct computational secret-sharing schemes for composed gap access structures based on the hardness of the learning parity with noise (LPN) problem. To present the ideas of our construction, we consider in the introduction an access structure $\text{Gap}_{\alpha, \beta}(\Gamma_1, \dots, \Gamma_m)$, where each access structure Γ_i is an arbitrary access structure with $0.1 \log m$ parties.

A simple way to construct a scheme for this access structure is to first share the secret in a ramp secret-sharing scheme with m parties such that any set of at least βm parties can reconstruct the secret and any set of less than αm parties learns no information on the secret, and then share the i -th share (for $1 \leq i \leq m$) independently using a perfect secret-sharing scheme realizing Γ_i . The best known implementation of this

scheme for the above access structures, using the perfect secret-sharing scheme of [4], will result in a secret-sharing scheme with share size $m^{0.058}$.

We construct a computational secret-sharing scheme for the gap access structure with share size $O(\log m)$; this scheme is secure under the LPN assumption. Our scheme uses a random matrix C over $\mathbb{F}_2^{cm \times m}$ (for some constant $c < 1$) to implement the ramp secret-sharing scheme, i.e., the shares are generated by sampling a random $x \in \mathbb{F}_2^{ck}$ and computing the shares as Cx ; by [19] this is a ramp secret-sharing scheme. We now share the i -th share using a $\frac{1}{m^2}$ -WP secret-sharing scheme. We show that the uncertainty of the secret in a WP secret-sharing scheme implies the computational security of this scheme. The technical challenge in the security proof is to take a noisy vector $y = Cx + e$, where each coordinate of e is independently sampled using the Bernoulli distribution with parameter $\frac{1}{m^2}$, and to construct shares of non-authorized sets for the vector Cx in the WP secret-sharing schemes for $\Gamma_1, \dots, \Gamma_m$. This is done by using a lemma about differential privacy [33] that states that the output of a differentially-private algorithm $\mathcal{A}(z)$ with a binary input z can be simulated as $g(y)$, where $y = z + e$ for e sampled from a Bernoulli distribution.

1.2.3 Weakly Private Decomposable Randomized Encoding and MPC

A construction of WP DRE for all functions with communication $\tilde{O}(n^2)$. We describe below how to construct a WP decomposable randomized encoding (DRE) for an arbitrary function $f : \{0, 1\}^n \rightarrow \{0, 1\}$. The idea of our construction is to design a mechanism g that reveals the input x almost all the time, but with some (small) probability reveals a random input x' , under the condition that $f(x') = f(x)$. Intuitively, the output of this mechanism satisfies our desired WP property as, given only the output of the mechanism, one cannot be sure whether it is the actual input x or some randomly sampled x' satisfying $f(x') = f(x)$. We then implement such a mechanism g securely via an existing DRE scheme with standard security from [35]; as g is a fairly simple function, it has an efficient DRE with standard security. Intuitively, this ensures that the final encoding reveals only the mechanism output, which satisfies our WP requirement.

In more detail, our WP encoding function $\text{ENC}(x)$ is described below. We first sample a uniformly distributed string $w \in \{0, 1\}^n$ (where w is a guess of the input x) and then choose a uniformly distributed string $w' \in \{0, 1\}^n$ such that $f(w) = f(w')$. The WP decomposable randomized encoding is computed as follows:

- Run a standard DRE encoding for the function $g(x, w, w')$ that outputs w' if the guess is correct, i.e. indeed $x = w$, and outputs x otherwise. Then, output the encoding of the function $g(x, w, w')$.

The WP decoding function Dec runs the standard DRE decoding algorithm to recover either the original x or the randomly sampled w' . It then outputs the evaluation of f over the recovered output of g ; i.e., x or w' . Note that the evaluation result is correct in either case of $f(x)$ or $f(w') = f(w = x)$.

Notably, we only apply a fully secure DRE on a simple function g whose complexity depends only on the input length n , and not on the function f . Plugging in an existing DRE construction from [35], we obtain a construction with size $\tilde{O}(n^2)$. We show that the construction satisfies p -WP for $p = 2^{-O(n)}$ in Section 5.1.

An almost matching lower bound. In Section 5.1.1, we show that the lower bound of $\Omega(n^2 / \log n)$ for DREs in [6] still holds even assuming only p -WP privacy, for some positive $p > 0$. We outline the arguments here of a weaker result, that is, there is an explicit function g such that in any DRE of g there exists a j such that the size of the encoding ENC_j is $\Omega(n / \log n)$.

Consider the function $g : \{0, 1\}^{n+\log n} \rightarrow \{0, 1\}$, where we parse the input of g as an n bit string $x = (x_1, \dots, x_n)$ and an index $i \in [n]$ (this is a $\log n$ -bit string) and define $g(x, i) = x_i$. Let ENC be a DRE of g .

Fix an input $x \neq 0^n, 1^n$ and a random string r ; for $1 \leq j \leq n$ let $\hat{y}_j = \text{ENC}_j(x_j; r)$ (that is, $\hat{y}_j$ is the local encoding of x_j with r). Let $z \in \{0, 1\}^n$ be any string. By the choice of x , there is some i such that $x_i = z_1$; thus, $g(x, i) = g(z, 1)$. The weak privacy implies that the local encodings $\hat{y}_1, \dots, \hat{y}_n$ are possible given z , that is, there is some r_z such that $\hat{y}_j = \text{ENC}_j(z_j; r_z)$ for $1 \leq j \leq n$. To conclude, $\hat{y}_1, \dots, \hat{y}_n$ (without the encodings of i) do not rule out any n -bit string.

We use the above facts to describe z by the encoding $\text{ENC}_{n+\ell}$ for $1 \leq \ell \leq \log n$ as follows (forgetting any privacy requirements). For every $z \in \{0, 1\}^n$, fix r_z as above; for $1 \leq \ell \leq \log n$ and $b \in \{0, 1\}$, let $\hat{y}_{\ell,b} = \text{ENC}_{\ell+n}(b, r_z)$ (that is, we consider all possibilities of the encoding of the bits of an index $i = (i_1, \dots, i_{\log(n)})$ with the string r_z). The correctness requirement implies that for every $i = (i_1, \dots, i_{\log(n)})$ given $\hat{y}_j$ for $1 \leq j \leq n$ and $\langle \hat{y}_{\ell, i_b} \rangle_{1 \leq \ell \leq \log n}$ one can recover z_i . In other words, given the fixed encodings $\langle \hat{y}_j \rangle_{1 \leq j \leq n}$ and the $2 \log n$ encodings $\langle \hat{y}_{\ell,b} \rangle_{1 \leq \ell \leq \log n, b \in \{0,1\}}$ one can recover the entire string z . This implies that for at least one z the size of $\langle \hat{y}_{\ell,b} \rangle_{1 \leq \ell \leq \log n, b \in \{0,1\}}$ is at least n ; i.e., there is at least one ℓ such that the size of $\text{ENC}_{n+\ell}$ is at least $n/2 \log n$.

A construction of WP MPC protocols for all functions. In this paper, we construct WP MPC protocols secure against semi-honest parties in two models: (1) protocols assuming an honest majority in the plain model, and (2) protocols secure against any number of semi-honest parties in the OT-hybrid model. The idea of the protocols is similar to the DRE construction, where to securely compute $f : (\{0, 1\}^n)^k \rightarrow \{0, 1\}$ the parties reveal their input $x_1, \dots, x_k$ almost all the time; however, with some (small) probability they reveal a random input $y_1, \dots, y_k$, under the condition that $f(x_1, \dots, x_k) = f(y_1, \dots, y_k)$. To implement this idea, we define a simple functionality g and use known perfect MPC protocols of [14, 30] to compute g .

To be concrete, we consider here the two-party case and the OT-hybrid model (all cases are detailed in Sections 5.2 and 5.3); we denote the two parties by Alice and Bob. Given a function $f : (\{0, 1\}^n)^2 \rightarrow \{0, 1\}$, we define the function $g : (\{0, 1\}^n)^4 \times \{0, 1\}^2 \rightarrow (\{0, 1\}^n)^2$, where $g((x_0, x_1^0, y_0^0, y_1^0, r_0), (x_1, x_0^1, y_0^1, y_1^1, r_1))$ is computed as follows:

- If $r_0 = r_1$ (where $r_0, r_1 \in \{0, 1\}$), $x_1^0 = x_1$, and $f(x_0, x_1^0) = f(y_0^0, y_1^0)$, then output y_0^0, y_1^0 .
- If $r_0 \neq r_1$, $x_0^1 = x_0$, and $f(x_0^1, x_1) = f(y_0^1, y_1^1)$, then output y_0^1, y_1^1 .
- Otherwise, output x_0, x_1 .

The protocol for computing $f(x_0, x_1)$ is as follows: Alice (resp. Bob) samples with a uniform distribution $x_1^0, y_0^0, y_1^0 \in \{0, 1\}^n$ and $r_0 \in \{0, 1\}$ (resp., $x_0^1, y_0^1, y_1^1 \in \{0, 1\}^n$ and $r_1 \in \{0, 1\}$). Thereafter, the parties execute a secure 2-party protocol in the OT-hybrid model computing $g((x_0, x_1^0, y_0^0, y_1^0, r_0), (x_1, x_0^1, y_0^1, y_1^1, r_1))$, get the output z_0, z_1 of g , and output $f(z_0, z_1)$.

To understand the function g , the inputs x_0, x_1 are the true inputs of the parties (i.e., the parties want to compute $f(x_0, x_1)$), the input x_1^0 is Alice's guess of Bob's input, and y_0^0, y_1^0 are the random inputs that (with small probability) will replace the true inputs; the condition $f(x_0, x_1^0) = f(y_0^0, y_1^0)$ (resp., $f(x_0^1, x_1) = f(y_0^1, y_1^1)$) ensures the correctness of the protocol. The inputs r_0, r_1 provide the privacy. Namely, to prove the weak privacy of the view of Bob, we consider inputs x_0, x'_0, x_1 such that $f(x_0, x_1) = f(x'_0, x_1)$. The most likely output of g when Alice and Bob hold x_0, x_1 is x_0, x_1 . However, x_0, x_1 can also be the output of g as seen by Bob when Alice and Bob hold x'_0, x_1 ; if Alice and Bob choose random inputs such that $r_0 = r_1$, $x_1^0 = x_1$, $y_0^0 = x_0$, and $y_1^0 = x_1$, then $g((x'_0, x_1, x_0, x_1, r_0), (x_1, x_1^0, y_0^0, y_1^0, r_0)) = (x_0, x_1)$.

We next explain how to construct a perfect 2-PC for g . First note that Alice (resp., Bob) can locally check if $f(x_0, x_1^0) = f(y_0^0, y_1^0)$ (resp., $f(x_0^1, x_1) = f(y_0^1, y_1^1)$). Given this information, the function g can be

computed by a Boolean circuit of size $O(n)$ with gates of fan-in 2; thus, the GMW protocol [30] can be used to compute g in the OT-hybrid model, using $O(n)$ calls to OT.

2 Preliminaries

Notation. Given a finite set A , we use the notation $y \leftarrow A$ to denote assigning y a value chosen uniformly at random from A . Similarly, for a randomized algorithm Alg , we write $y \leftarrow \text{Alg}(x)$ to denote assigning y the output of Alg on input x . The *binary entropy function* is $h : [0, 1] \rightarrow [0, 1]$, defined as $h(x) = -x \log x - (1 - x) \log(1 - x)$ for all $x \in (0, 1)$, with $h(0) = h(1) = 0$.

2.1 Information-Theoretic Secret-Sharing Schemes

In this section, we present definitions related to Information-Theoretic (IT) Secret-Sharing Schemes.

Definition 2.1 (Partial access structures). *Let $P = \{p_1, \dots, p_n\}$ be a set of parties. A partial access structure is a pair of collections of sets $\Gamma = (\Gamma_{\text{no}}, \Gamma_{\text{yes}})$, where $\Gamma_{\text{no}}, \Gamma_{\text{yes}} \subseteq 2^P$ such that $\emptyset \notin \Gamma_{\text{yes}}$ and $B \not\subseteq A$ for every $A \in \Gamma_{\text{no}}, B \in \Gamma_{\text{yes}}$. Sets in Γ_{yes} are called authorized and sets in Γ_{no} are called unauthorized. If $\Gamma_{\text{no}} \cup \Gamma_{\text{yes}} = 2^P$, we say that Γ is a complete access structure, or just an access structure, and will denote it by $\Gamma = \Gamma_{\text{yes}}$.*

We only have a minimal consistency requirement on partial access structures, i.e., an authorized set in Γ_{yes} will be able to reconstruct the secret; thus, it cannot be contained in an unauthorized set in Γ_{no} that should not get “too much” information on the secret.

A secret-sharing scheme is a randomized mapping that, given a secret s , outputs n strings called shares such that any authorized set of parties can reconstruct the secret from its shares. We consider a security requirement that we call weakly private, denoted WP; a secret-sharing scheme is p -WP (for $p \in [0, 1]$) if, for any unauthorized set A , the probability of any shares of A given one secret is at least p times the probability of those shares given any other secret.

Definition 2.2 (Weakly private (WP) secret-sharing schemes). *A secret-sharing scheme with domain of secrets S such that $|S| \geq 2$ and domains of shares $S_1, S_2, \dots, S_n$ is a randomized mapping Π that takes as input a secret $s \in S$ and outputs an n -tuple $\langle \text{sh}_1, \dots, \text{sh}_n \rangle \in S_1 \times S_2 \times \dots \times S_n$. For a set $A \subseteq P$, we denote $\Pi_A(s)$ as the restriction of $\Pi(s)$ to its A -entries (i.e., the shares of the parties in A ; namely, $\langle \text{sh}_i \rangle_{p_i \in A}$).*

Given a secret-sharing scheme Π , define the size of the secret as $\log |S|$, the share size of party p_j as $\log |S_j|$, the max share size as $\max_{1 \leq j \leq n} \{\log |S_j|\}$, and the total share size as $\sum_{j=1}^n \log |S_j|$.

Let $0 < p \leq 1$. We say that Π is a p -WP secret-sharing scheme realizing an access structure $\Gamma = (\Gamma_{\text{no}}, \Gamma_{\text{yes}})$ if the following two requirements hold:

Perfect correctness. *The secret s can be reconstructed by any authorized set of parties. That is, for any set $B = \{p_{i_1}, \dots, p_{i_{|B|}}\} \in \Gamma_{\text{yes}}$ there exists a reconstruction function $\text{Recon}_B : S_{i_1} \times \dots \times S_{i_{|B|}} \rightarrow S$ such that $\text{Recon}_B(\Pi_B(s)) = s$, for every secret $s \in S$ and every randomness of Π .*

p -WP. *Every unauthorized set cannot learn “too much” information about the secret from its shares. Formally, for any set $A = \{p_{i_1}, \dots, p_{i_{|A|}}\} \in \Gamma_{\text{no}}$, every two secrets $s_1 \neq s_2 \in S$, and every vector of shares $\langle \text{sh}_{i_1}, \dots, \text{sh}_{i_{|A|}} \rangle \in S_{i_1} \times \dots \times S_{i_{|A|}}$,*

$$\Pr[\Pi_A(s_1) = \langle s_j \rangle_{p_j \in A}] \geq p \cdot \Pr[\Pi_A(s_2) = \langle s_j \rangle_{p_j \in A}].$$

A secret-sharing scheme is a perfect secret-sharing scheme if it is 1-WP. A secret-sharing scheme is a weakly private secret-sharing scheme if it is p -WP for some $p > 0$.

Our definition of privacy is inspired by the definition of (pure) ε -differential privacy, a notion introduced by Dwork, McSherry, Nissim, and Smith [25]; since we deal with very weak privacy, we use a different parameterization; namely, p -WP corresponds to $\varepsilon = \ln(1/p)$.

2.2 Computational and Statistical Secret-Sharing Schemes

In this section we define *computational* secret-sharing schemes (resp., *statistical* secret-sharing schemes), that is, secret-sharing schemes in which the sharing and reconstruction are efficient, and no polynomial-time adversary (resp., any unbounded adversary) can learn non-negligible information about the secret from the shares of any unauthorized set of parties. When defining “efficiency”, it is important to consider the way the access structure is represented. Following [1], we start with an abstract definition of a representation model for access structures.

Definition 2.3 (Representation model). A representation model is a polynomial time computable function $U : \{0, 1\}^* \times 2^{\{p_1, p_2, \dots\}} \rightarrow \{0, 1, *\}$, where the input of U is a “program” Prog and a set A ; Prog represents a partial access structure as follows: if $U(\text{Prog}, A) = 1$ then A is authorized, if $U(\text{Prog}, A) = 0$ then A is unauthorized, and if $U(\text{Prog}, C) = *$, there are no requirements for C . We assume that each Prog specifies the number of parties n in the access structure (i.e., $U(\text{Prog}, C) = *$ if $C \not\subseteq \{p_1, \dots, p_n\}$) and assume that $|\text{Prog}| \geq n$.

We now define computational secret-sharing schemes. For simplicity, we assume here that the secret is a single bit. This can always be generalized to sharing an ℓ -bit secret by independently sharing each bit of the secret. Moreover, using a symmetric encryption scheme, the amortized cost of sharing long secrets can be improved.

Definition 2.4 (Computational secret sharing – Syntax and correctness). A computational secret-sharing (CSS) scheme for a representation model U consists of a pair of algorithms $\text{CSS} = (\text{CSS.SHARE}, \text{CSS.RECON})$ with the following syntax.

Sharing. $\text{CSS.SHARE}(\text{Prog}, s) \rightarrow (\text{sh}_0, \text{sh}_1, \dots, \text{sh}_n)$ (where n denotes the number of parties in Prog) is a randomized poly-time algorithm that takes as input a program Prog and a secret $s \in \{0, 1\}$ and outputs $n + 1$ shares $\text{sh}_0, \text{sh}_1, \dots, \text{sh}_n$, where sh_0 is public information and sh_i , for $1 \leq i \leq n$, is the share of party p_i .

Reconstruction. $\text{CSS.RECON}(\text{Prog}, A, \text{sh}_0, (\text{sh}_i)_{p_i \in A}) \rightarrow s$ is a deterministic poly-time algorithm that takes as input a program Prog, a set $A \subseteq \{p_1, \dots, p_n\}$ (where n denotes the number of parties in Prog), the public share, and the shares of the parties in A . The algorithm outputs a value $s \in \{0, 1\}$.

We say that CSS is correct (with respect to U) if there is a negligible function $\varepsilon(n)$ such that for every program Prog and a set $A \subseteq \{p_1, \dots, p_n\}$ such that $\text{Prog}(A) = 1$ (where n denotes the number of parties in Prog), the process of invoking $\text{CSS.SHARE}(\text{Prog}, s)$ and then invoking $\text{CSS.RECON}(\text{Prog}, A, \text{sh}_0, (\text{sh}_i)_{p_i \in A})$ (with the shares sh_i generated by CSS.SHARE) returns s with probability at least $1 - \varepsilon(n)$.

Definition 2.5 (Computational and statistical secret-sharing schemes – Security). Consider the following game between a non-uniform poly-time adversary $\mathcal{A}$ and a challenger:

1. The adversary $\mathcal{A}$ on input 1^n chooses a program Prog , representing an n -party access structure, and a set $A \subseteq \{p_1, \dots, p_n\}$ such that $\text{Prog}(A) = 0$ and sends them to the challenger.
2. The challenger chooses a secret bit $s \leftarrow_U \{0, 1\}$ uniformly at random. It computes $(\text{sh}_0, \text{sh}_1, \dots, \text{sh}_n) \leftarrow \text{CSS.SHARE}(\text{Prog}, s)$ and sends $\text{sh}_0, (\text{sh}_i)_{p_i \in A}$ to the adversary.
3. The adversary outputs a bit s' .

The adversary wins the game if $s' = s$.

We say that CSS is computationally secure if, for every non-uniform poly-time adversary $\mathcal{A}$, there exists a negligible function $\varepsilon(n)$ such that the probability that $\mathcal{A}$ wins the above game is at most $1/2 + \varepsilon(n)$. We say that CSS is statistically secure if the above holds for every unbounded adversary $\mathcal{A}$.

2.3 Useful Access Structures

We will use the following access structures in the sequel.

Definition 2.6 (Forbidden graph access structures). Given an undirected graph $G = (V, E)$, define the partial access structure $\Gamma_G = \langle \Gamma_{G,\text{no}}, \Gamma_{G,\text{yes}} \rangle$ whose parties are V and the collections are $\Gamma_{G,\text{no}} = \{\{u, v\} : (u, v) \notin E\}$ and $\Gamma_{G,\text{yes}} = \{\{u, v\} : (u, v) \in E\}$, that is, each set of size 2 is either authorized or unauthorized; it is authorized if and only if it is an edge in the graph. There are no additional requirements for sets of size larger than 2.

Ramp secret-sharing schemes generalize threshold secret-sharing schemes. In a t -private, k -correct ramp secret-sharing scheme, sets of parties of size at least k should be able to reconstruct the secret, while sets of parties of size at most t should get no information on the secret. There are no requirements on sets with more than k parties but less than t parties. Allowing a gap between t and k results in schemes that are more efficient than threshold secret-sharing schemes. Ramp secret-sharing schemes were first presented by Blakley and Meadows [16], and were used to construct efficient secure multiparty computation (MPC) protocols, starting in the work of Franklin and Yung [28].

Definition 2.7 (Ramp secret-sharing schemes). Let $0 \leq t < k \leq n$. A secret-sharing scheme is a t -private, k -correct ramp secret-sharing scheme if it realizes the following partial access structure $\Gamma_{\text{ramp}} = (\Gamma_{\text{ramp,no}}, \Gamma_{\text{ramp,yes}})$, where $\Gamma_{\text{ramp,no}} = \{A \subset \{p_1, \dots, p_n\} : |A| \leq t\}$ and $\Gamma_{\text{ramp,yes}} = \{A \subset \{p_1, \dots, p_n\} : |A| \geq k\}$.

2.4 Binary Symmetric Channels (BSC) and WP Secret-Sharing Schemes

A δ -BSC channel is a randomized mapping with domain $\{0, 1\}$ that for every $s \in \{0, 1\}$ outputs $\bar{s}$ with probability δ and outputs s with probability $1 - \delta$; i.e., with probability δ the value of the bit is flipped. The following lemma is a special case of a result of [33]; this lemma asserts that a p -WP secret-sharing scheme with a binary secret can be simulated from a BSC channel whose input is a secret. For completeness, we provide the proof of the lemma.

Lemma 2.8. Let Π be a randomized mapping with a domain $\{0, 1\}$ and range O , and let $0 \leq p \leq 1$. If, for every $o \in O$ and every $s \in \{0, 1\}$,

$$\Pr[\Pi(s) = o] \geq p \cdot \Pr[\Pi(\bar{s}) = o], \quad (1)$$

then Π can be simulated from a δ -BSC channel $\text{BSC}(s)$, for $\delta = \frac{p}{1+p}$. That is, there is a randomized mapping g with domain $\{0, 1\}$ such that, for every $o \in O$ and every $s \in \{0, 1\}$,

$$\Pr[\Pi(s) = o] = \Pr[g(\text{BSC}(s)) = o]. \quad (2)$$

Furthermore, if for every $s \in \{0, 1\}$ and $o \in O$, the probability $\Pr[\Pi(s) = o]$ can be efficiently computed, then g can be efficiently computed.

Proof. Let $p_{s,o} = \Pr[\Pi(s) = o]$. The mapping g on input $s' \in \{0, 1\}$ outputs o with probability

$$q_{s',o} = \frac{1}{1-p} (p_{s',o} - p \cdot p_{\bar{s}',0}).^5$$

We first prove that $\langle q_{s',o} \rangle_{o \in O}$ is a probability distribution for every $s' \in \{0, 1\}$. By (1), $q_{s',o} \geq 0$ for every $o \in O$. Furthermore, for every $s' \in \{0, 1\}$,

$$\sum_{o \in O} q_{s',o} = \frac{1}{1-p} \sum_{o \in O} (p_{s',o} - p \cdot p_{\bar{s}',0}) = \frac{1}{1-p} \left(\sum_{o \in O} p_{s',o} - p \cdot \sum_{o \in O} p_{\bar{s}',0} \right) = \frac{1}{1-p} (1 - p) = 1.$$

Finally, we prove that (2) holds. Note that $\Pr[\text{BSC}(s) = \bar{s}] = \frac{p}{1+p}$ and $\Pr[\text{BSC}(s) = s] = 1 - \frac{p}{1+p} = \frac{1}{1+p}$. Thus,

$$\begin{aligned} \Pr[g(\text{BSC}(s)) = o] &= \Pr[\text{BSC}(s) = s] \cdot \Pr[g(s) = o] + \Pr[\text{BSC}(s) = \bar{s}] \cdot \Pr[g(\bar{s}) = o] \\ &= \frac{1}{1+p} \cdot \frac{1}{1-p} (p_{s,o} - p \cdot p_{\bar{s},0}) + \frac{p}{1+p} \cdot \frac{1}{1-p} (p_{\bar{s},o} - p \cdot p_{s,0}) \\ &= \frac{1-p^2}{(1+p)(1-p)} p_{s,o} = p_{s,o}. \end{aligned}$$

□

2.5 LPN and Random Linear Error-Correcting Codes

Our computational secret-sharing schemes are based on the hardness of the learning parity with errors (LPN) problem; i.e., a noisy encoding of a random string using a random linear error-correcting code is indistinguishable from a random string. We next provide some background on random linear error-correcting codes and the LPN problem.

Given an $[m+1, k, d]$ linear error-correcting code C over a finite field $\mathcal{K}$, define the m -party secret-sharing scheme $\text{LSSS}(C)$, where to share a secret $s \in \mathcal{K}$, one chooses a random codeword $y = (y_0, \dots, y_m)$ such that $y_0 = s$ and the share of party p_i is y_i .

Theorem 2.9 ([19]). *Let C be an $[m+1, k, d]$ -code over a finite field $\mathcal{K}$ whose dual code has distance $d^\perp$. Then $\text{LSSS}(C)$ is a linear $(d^\perp - 2)$ -private, $(m - d + 2)$ -correct ramp secret-sharing scheme.*

Theorem 2.10 ([19]). *Let $0 < \delta < 1/2$. The distance of a random linear code over $\mathbb{F}_2$ of length m and dimension k is at least δm except with probability $2^{k-m(1-h(\delta))}$.*

In the LPN problem, a random linear error-correcting code $C \in \mathbb{F}_2^{m \times k}$ and an information word x are uniformly and independently sampled, a noisy codeword $y = (y_1, \dots, y_m)$ is computed as $y = Cx + e$, where each coordinate of e is sampled from a Bernoulli distribution. The LPN assumption assumes that the

⁵This expression is chosen such that (2) holds, i.e., for every $o \in O$ the values of $q_{0,o}, q_{1,o}$ is the solution to the following linear equations for $s = 0, 1$

$$p_{s,o} = \Pr[\text{BSC}(s) = s] \cdot \Pr[g(s) = o] + \Pr[\text{BSC}(s) = \bar{s}] \cdot \Pr[g(\bar{s}) = o] = (1 - \delta) \cdot q_{s,o} + \delta \cdot q_{\bar{s},o}.$$

distribution C, y is indistinguishable from the distribution C, u , where u is a uniformly distributed string. For the security of our computational secret-sharing schemes, we need a version of the LPN problem in which the first $\alpha \cdot m$ coordinates of the noise vector e are 0, that is, we consider a version of the LPN problem in which the first αm coordinates of the codeword are exposed without noise. We show that this version is equivalent to the standard LPN assumption where there is no exposure.

Definition 2.11 (Learning parity with noise (LPN) with known coordinates). *Let k be the dimension, $m(k)$ be the number of samples, i.e., the length of the code word, $\mu(k)$ be the error rate, and α be the exposure parameter (where $k, m(k) \in \mathbb{N}$ and $0 \leq \mu(k), \alpha < 1$). The learning parity with noise assumption $\text{LPN}(m(k), \mu(k), \alpha)$ assumes the following distributions to be computationally indistinguishable*

$$\{(C, Cx + e)\}_k \approx_c \{(C, u)\}_k,$$

where the public matrix C , the random word u , and the secret vector x are sampled at random as $C \leftarrow \mathbb{F}_2^{m(k) \times k}$, $u \leftarrow \mathbb{F}_2^{m(k)}$, $x \leftarrow \mathbb{F}_2^k$, and the error vector $e = (e_1, \dots, e_m)$ is computed such that $e_1 = \dots = e_{\lfloor \alpha \cdot m(k) \rfloor} = 0$ and each e_i , for $\lfloor \alpha \cdot m(k) \rfloor + 1 \leq i \leq m(k)$, is independently sampled from a Bernoulli distribution with parameter $\mu(k)$. We denote $\text{LPN}(m(k), \mu(k), 0)$ by $\text{LPN}(m(k), \mu(k))$.

It was observed by Dodis et al. [23] (see comment on Page 5) that the $\text{LPN}(m(k), \mu(k), \alpha)$ assumption is implied by the LPN assumption without exposure, i.e., the standard LPN assumption (with smaller parameters as stated in the next lemma). In particular, $\text{LPN}(m(k) = O(k), \mu = 1/\Omega(\sqrt{k}))$ implies $\text{LPN}(m(k) = ck, \mu = 1/\Omega(\sqrt{k}), \alpha)$ for any constants $c \cdot \alpha < 1$. This follows from the fact that linear leakage of the LPN secret is equivalent to leaking bits of the secret, which effectively reduces its secret dimension. For completeness, we prove the lemma below.

Lemma 2.12. *Let c, α be constants such that $c > 1$ and $\alpha c < 1$ and let $c' = (1 - \alpha)c/(1 - \alpha c)$. If the $\text{LPN}(c'k, \mu(k/(1 - \alpha c)))$ assumption holds, then the $\text{LPN}(ck, \mu(k), \alpha)$ assumption holds.*

Proof. We describe an efficient transformation that, given a random input to the standard LPN problem, constructs a random input for the LPN assumption with exposure. That is, the transformation for a given k gets an input $C' \in \mathbb{F}_2^{c'k' \times k'}$, $y' \in \mathbb{F}_2^{c'k'}$, where $k' = (1 - \alpha c)k$ and $c' = \frac{(1 - \alpha)c}{1 - \alpha c}$ and outputs $C \in \mathbb{F}_2^{ck \times k}$, $y \in \mathbb{F}_2^{ck}$ such that

1. C is almost uniformly distributed,
2. If y' is uniformly distributed then y is uniformly distributed and independent of C ,
3. If $y' = C'x' + e'$ where x' is uniformly distributed and the each coordinate of e' is sampled from a Bernoulli distribution with parameter $\mu(k)$, then $y = Cx + e$ where x is uniformly distributed, the first αm coordinates of e are zero and each other coordinate is sampled from a Bernoulli distribution with parameter $\mu(k)$.

Thus, if there is an adversary that breaks the LPN assumption with exposure, then there is an adversary that breaks the standard LPN assumption. In Figure 2, we spell out a transformation (the existence of matrices used in this transformation is proved below).

To complete the proof of the lemma, we prove that the matrix B in the transformation exists and that the above properties (1)-(3) hold. First note that since $\alpha m = \alpha ck < k$, a uniformly distributed matrix in $\mathbb{F}_2^{(\alpha m) \times k}$ has full row rank with probability $1 - 0.5^{O(k)}$. Thus, the statistical distance between $C_\top$ and a random $(\alpha m) \times k$ matrix is negligible. Second, an invertible matrix B such that $C_\top \cdot B = (I_{\alpha m} \| 0)$ exists – take the first αm

For an integer k , let $m = ck$, $k' = (1 - \alpha c)k = k - \alpha m$, and $m' = (1 - \alpha)ck = \frac{(1-\alpha)c}{1-\alpha c}k'$.

Input: $C' \in \mathbb{F}_2^{m' \times k'}$, $y' \in \mathbb{F}_2^{m'}$.

Output: $C \in \mathbb{F}_2^{m \times k}$, $y \in \mathbb{F}_2^m$.

1. Sample a random full-rank matrix $C_\top \leftarrow \mathbb{F}_2^{(\alpha m) \times k}$ and a random vector $y_\top \leftarrow \mathbb{F}_2^{(\alpha m)}$.
2. Compute an invertible matrix $B \in \mathbb{F}_2^{k \times k}$ such that $C_\top \cdot B = (I_{\alpha m} \| 0)$.
3. Sample $R \leftarrow \mathbb{F}_2^{m' \times \alpha m}$ and compute $C_\perp = (R \| C') \cdot B^{-1} \in \mathbb{F}_2^{m' \times k}$ and $y_\perp = y' + R \cdot y_\top \in \mathbb{F}_2^{k'}$.
4. Output

$$C = \begin{pmatrix} C_\top \\ C_\perp \end{pmatrix}, y = \begin{pmatrix} y_\top \\ y_\perp \end{pmatrix}.$$

Figure 2: A transformation from $\text{LPN}(c'k, \mu(k/(1 - \alpha c)))$ to $\text{LPN}(ck, \mu(k), \alpha)$.

rows of B^{-1} as $C_\top$ and arbitrarily complete it to a full-rank matrix with k rows. Third, since $(R \| C')$ is a random matrix and B^{-1} has full rank, the matrix $C_\perp$ is uniformly distributed. The above facts imply that the statistical distance between C and a random $m \times k$ is negligible.

The claim that random samples y' are transformed to random outputs y independent of C is clear. In the case where $y' = C'x' + e'$, we show that the transformed samples satisfy $y_\top = C_\top x$ and $y_\perp = C_\perp x + e'$, where $x = B(y_\top^\top \| x'^\top)^T$:

$$C_\top x = C_\top (B(y_\top^\top \| x'^\top)^T) = (I_{\alpha m} \| 0)(y_\top^\top \| x'^\top)^T = y_\top$$

and

$$C_\perp x + e' = ((R \| C')B^{-1})(B(y_\top^\top \| x'^\top)^T) + e' = Ry_\top + C'x' + e' = y_\perp.$$

This completes the proof that the transformation satisfies the desired properties. $\square$

Remark 2.13. We can consider a version in which a set $I \subset [m]$ of size αm is chosen by an adversary prior to seeing C , and the noise vector e is chosen such that $e_i = 0$ for every $i \in I$ and each e_i for $i \notin I$ is independently sampled with a Bernoulli distribution with parameter $\mu(k)$. This version can be trivially transformed to the version of the problem discussed in Definition 2.11, where given I and C , y one permutes the rows of C and y such that the I -rows in C are permuted to the first αm rows.

2.6 Decomposable Randomized Encodings

We next define decomposable randomized encoding schemes, also known as garbling schemes and private simultaneous messages protocols. We take the definition from [6] and extend it with our weak privacy requirement.

Definition 2.14 (Decomposable randomized encodings). *A randomized encoding for a function $f : \{0, 1\}^n \rightarrow Y$ consists of a “randomness” distribution $\mathcal{R}$, an encoding algorithm $\text{ENC} : \{0, 1\}^n \times \mathcal{R} \rightarrow \{0, 1\}^\ell$, and a decoding algorithm $\text{DEC} : \{0, 1\}^\ell \rightarrow Y$, where ℓ is called the size of the randomized encoding.*

Perfect correctness. *The value of $f(x)$ can be computed from the encoding. That is, for any input $x \in \{0, 1\}^n$, for every $r \in \mathcal{R}$*

$$\text{DEC}(\text{ENC}(x; r)) = f(x).$$

p -WP. For any inputs $x_1, x_2 \in \{0, 1\}^n$ such that $f(x_1) = f(x_2)$, and any encoding $\widehat{y} \in \{0, 1\}^\ell$,

$$\Pr_{r \leftarrow \mathcal{R}} [\text{ENC}(x_1; r) = \widehat{y}] \geq p \cdot \Pr_{r \leftarrow \mathcal{R}} [\text{ENC}(x_2; r) = \widehat{y}].$$

Decomposability. Each output bit of $\text{ENC}(x; r)$ is determined by r and one bit of x ; that is, there are functions $\text{ENC}_1, \dots, \text{ENC}_n : \{0, 1\} \times \mathcal{R} \rightarrow \{0, 1\}^\ell$ such that for every $x = \langle x_1, \dots, x_n \rangle \in \{0, 1\}^n$ and every $r \in \mathcal{R}$,

$$\text{ENC}(x; r) = \langle \text{ENC}_1(x_1; r), \dots, \text{ENC}_n(x_n; r) \rangle.$$

The size of a decomposable randomized encoding is

$$\max_{1 \leq i \leq n, b \in \{0, 1\}, r \in \mathcal{R}} |\text{ENC}_i(b; r)|.$$

A WP decomposable randomized encoding is a p -WP decomposable randomized encoding for some positive $p > 0$, and a (standard) decomposable randomized encoding is a 1-WP decomposable randomized encoding.

Note that in a DRE, all local encoding functions use the same random string r . Efficient constructions of decomposable randomized encoding were given in [31, 10, 11, 35, 5]; we will use the following result of Kolesnikov [35].

Lemma 2.15 (Decomposable randomized encoding for boolean formulas [35]). *There exists a decomposable randomized encoding for any Boolean formula $f : \{0, 1\}^n \rightarrow \{0, 1\}$ with size $O(2^d d^2 \log d)$, where d is the depth of f .*

2.7 WP MPC Protocols

We consider k -party protocols for semi-honest parties both in the plain model and in the OT-hybrid model. In the plain model, the k parties $\{1, \dots, k\}$ exchange messages on secure and private point-to-point channels. In the OT-hybrid model, in addition, in each step of the protocol, any ordered pair of parties can use an ideal blackbox that computes the oblivious transfer function, that is, the input of the first party is an index $i \in \{0, 1\}$, the input of the second party is two bits i_0, i_1 , the first party gets as an output i_b , and the second party gets no output. The view of a coalition of parties T in a protocol Π in the plain model, denoted $\text{View}_T(x_1, \dots, x_k)$, is a random variable consisting of the inputs of the parties in the set, their randomness, and the communication exchanged between the parties in the coalition and the parties not in the coalition. In the OT-hybrid mode, the view also contains the outputs of the OT blackboxes given to the parties in the coalition.

Definition 2.16 (k -party WP MPC protocols). *A k -party, t -secure, p -WP MPC protocol Π for a function $f : (\{0, 1\}^n)^k \rightarrow \{0, 1\}$ in the plain or OT-hybrid model is a k -party protocol satisfying the following two properties:*

Perfect correctness. *The output of Π always equals $f(x_1, \dots, x_k)$. That is, for any input $(x_1, \dots, x_k) \in (\{0, 1\}^n)^k$,*

$$\Pr_{r_1, \dots, r_k} [\Pi(x_1, \dots, x_k) = f(x_1, \dots, x_k)] = 1.$$

***p*-WP.** For every coalition $T \subseteq \{1, \dots, k\}$ such that $|T| \leq t$, every inputs $x_1, \dots, x_k, x'_1, \dots, x'_k$ such that $x_i = x'_i$ for every $i \in T$ and $f(x_1, \dots, x_k) = f(x'_1, \dots, x'_k)$, and every possible view view_T of the parties in T ,

$$\Pr_{(r_i)_{i \notin T}} [\text{View}_T(x_1, \dots, x_k) = \text{view}_T] \geq p \cdot \Pr_{(r_i)_{i \notin T}} [\text{View}_T(x'_1, \dots, x'_k) = \text{view}_T].$$

In general, MPC protocols are usually defined using the real vs. ideal paradigm (i.e., by using simulators). For deterministic functions, the definition of MPC protocols that are secure against unbounded, semi-honest parties in the real vs. ideal paradigm is equivalent to separately requiring correctness and privacy. Thus, in the above definition, in order to emphasize its weak-privacy nature, we separately define correctness and privacy (rather than using a simulator).

3 Constructions of WP Secret-Sharing Schemes

In this section, we present a WP secret-sharing scheme for an arbitrary access structure with share size $O(t) + q$, where t is the maximum size of an unauthorized set and q is the length of the secrets. As a warm-up, we describe two simple schemes with binary secrets for forbidden graphs (i.e., access structures in which every maximal unauthorized set has size 2). In the first scheme, correctness is perfect and the share size is 1; however, the scheme does not provide weak privacy. The second is $1/O(n^2)$ -WP; however, the correctness is not perfect.

3.1 Warm-up: A Simple Construction for Forbidden Graphs

We start with a very simple secret-sharing scheme for forbidden graphs Γ , described in Figure 3. This scheme has perfect correctness, but is *not* weakly private and it only offers some “average case” security: any unauthorized set of size 2 with a small probability ($\geq 1/n^2$ for an n -party access structure) will not get any information on the secret; however, in all other cases, this set will learn everything (i.e., the value of s). Although this scheme provides little security, it will be used in Section 4.1 to construct a secret-sharing scheme with statistical security for a composed gap access structure.

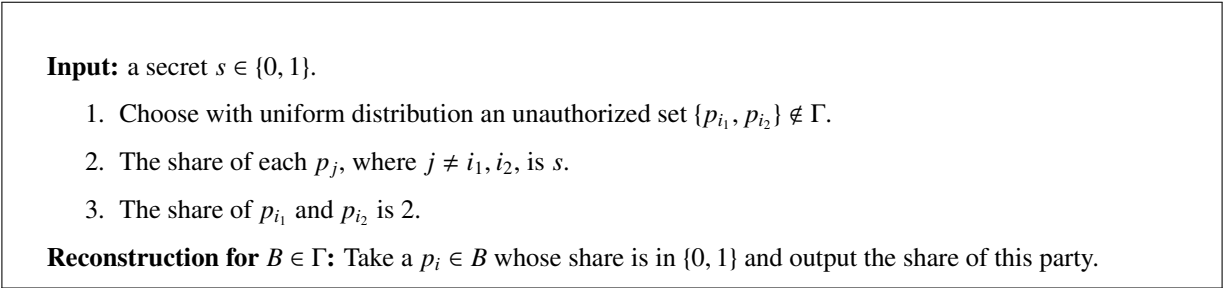


Figure 3: A simple non-private secret-sharing scheme for forbidden graphs.

We next explain the correctness and security of the secret-sharing scheme described in Figure 3. For the correctness, note that if the share of a party is a value in $\{0, 1\}$, then this value is the secret. Let B be an authorized set. Since $\{p_{i_1}, p_{i_2}\}$ is unauthorized, there is a party $p_i \in B \setminus \{p_{i_1}, p_{i_2}\}$; the share of such p_i is the secret s and this is the value reconstructed in the scheme. For the security, consider an unauthorized set $\{u, v\} \notin \Gamma$. With probability at least $1/n^2$, this set is chosen in Step 1; in this case, the parties get no information about the secret.

3.2 A WP Scheme for Forbidden Graphs with Imperfect Correctness

We next present in Figure 4 a $O(1/n^2)$ -WP secret-sharing scheme for forbidden graphs in which any authorized set reconstructs the correct secret with a high probability of at least $1 - 2/n$; with the remaining probability, it outputs $\perp$ (i.e., the reconstruction never errs). The advantage of this scheme is that the share of each party is a bit.

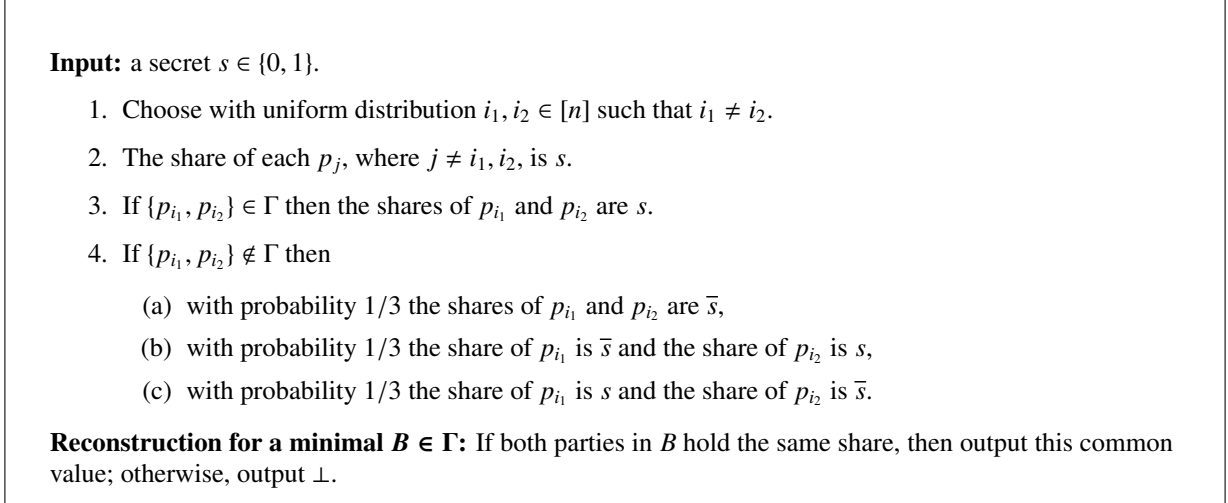


Figure 4: A $2/3n^2$ -WP secret-sharing scheme for forbidden graphs.

Lemma 3.1. *The scheme described in Figure 4 is a $2/3n^2$ -WP secret-sharing scheme with binary shares in which an authorized set reconstructs the correct secret with probability $1 - O(1/n)$ and with the remaining probability outputs $\perp$.*

Proof. We first prove the correctness and security of the scheme described in Figure 4. If the parties of an authorized set B hold the same share, then this is the correct secret s (since only an unauthorized set may hold the shares $\bar{s}, \bar{s}$). However, if the set B sees two different shares s, s' , then it cannot know which value is correct and outputs $\perp$; this event may occur only if one of the parties in B is in the set $\{p_{i_1}, p_{i_2}\}$ selected in Step 1, i.e., with probability at most $2/n$.

For every unauthorized set A , with probability $1/\binom{n}{2} > 2/n^2$ (where n is the number of parties in the access structure) this set is chosen in Step 1; in this case, each of the three pairs of shares $\langle \bar{s}, \bar{s} \rangle, \langle \bar{s}, s \rangle, \langle s, \bar{s} \rangle$ is selected for this set with probability $1/3$. Thus, regardless of the secret, every vector of shares in $\{0, 1\}^2$ has probability at least $2/3n^2$, in particular, the scheme is $2/3n^2$ -WP, i.e., for every $s \neq s' \in \{0, 1\}$ and every $s_1, s_2 \in \{0, 1\}$,

$$\Pr[\Pi_A(s) = \langle s_1, s_2 \rangle] \geq \frac{2}{n^2} \cdot \frac{1}{3} \geq \frac{2}{3n^2} \cdot \Pr[\Pi_A(s') = \langle s_1, s_2 \rangle].$$

□

3.3 A Construction with Perfect Reconstruction for Arbitrary Access Structures

Let Γ be an access structure in which all unauthorized sets are of size at most t . In Figure 5, we present a WP secret-sharing scheme realizing any such Γ (with perfect correctness). The scheme uses t -dominated graphs as defined in Definition 3.2.

Definition 3.2 (t -dominated graphs). Let $G = (V, E)$ be a directed graph without antiparallel edges and let $1 \leq t < |V|$ be an integer. We say that a vertex u dominates a set $W \subset V$ if $(u, v) \in E$ for every $v \in W$. We say that a directed graph $G = (V, E)$ is a t -dominated graph if for every $W \subset V$ such that $|W| = t$ there exists a vertex $u \in V \setminus W$ that dominates W .

For example, a directed cycle with 3 vertices is a 1-dominated graph, and in Figure 1 we describe a 2-dominated graph with 7 vertices. We can always assume that a t -dominated graph is a tournament (i.e., for every $u, v \in V$ either $(u, v) \in E$ or $(v, u) \in E$ (but not both)).

Remark 3.3. If a graph has a dominating set of size t , then it is not t -dominated. Recall that every tournament with N vertices has a dominating set of size $\log N$ (such a set can be constructed by the greedy algorithm using the fact that in every N -vertex tournament there is a vertex with out-degree $(N - 1)/2$). Thus, in every t -dominated graph, there are at least 2^t vertices.

In Figure 5, we describe a p -WP secret-sharing scheme for an arbitrary access structure that uses a t -dominated graph G , where t is the maximum size of an unauthorized set in the access structure. In this scheme to share a secret $s \in \{0, 1\}^q$, we choose a uniformly distributed maximal unauthorized set A (of size at most t); the share of each party in the set A is a random secret and a random vertex from G . We take some vertex u that dominates all the vertices in the shares of the parties in A and the share of each party not in A is $\langle s, u \rangle$, i.e., the secret that is shared and the dominating vertex. To achieve weak privacy, we can choose the vertex and secret of each party in A independently with uniform distribution. To achieve p -WP for a smaller p , we choose the vertices and secrets of the parties in A using a more involved process (see Procedure ChooseVertices in Figure 5). For example, for an arbitrary n -party access structure, the naive way of choosing vertices and secrets will result in a $1/2^{O(n^2)}$ -WP scheme, while using Procedure ChooseVertices results in a $1/2^{O(n)}$ -WP scheme.

Theorem 3.4 (WP secret sharing for general access structures). Let Γ be an access structure in which all unauthorized sets are of size at most t and $G = (V, E)$ be a t -dominated graph. Assume that Γ contains M maximal unauthorized sets. The scheme described in Figure 5 is a $\frac{1}{2M|V|^{2q+t}}$ -WP secret-sharing scheme realizing Γ with a q -bit secret and share size $\lceil \log(|V|) \rceil + q$.

Proof. We first prove the correctness of the scheme. Let $B \in \Gamma$ be an authorized set holding shares $\langle \langle s_j, v_j \rangle \rangle_{p_j \in B}$ and let A be the set chosen in Step 2. By the monotonicity of Γ , there exists $p_i \in B \setminus A$. Every party $p_i \in B \setminus A$ holds the share $\langle s, u \rangle$, where s is the correct secret and u dominates $\{v_j\}_{p_j \in A}$ – all vertices appearing in shares of A ; in particular, u dominates $\{v_j\}_{p_j \in B \cap A} = \{v_j\}_{p_j \in B} \setminus \{u\}$. Clearly, for every $p_j \in B \cap A$, the vertex v_j does not dominate u . Thus, a $p_i \in B \setminus A$ is chosen and the secret in its share is the correct secret.

We next prove the privacy of the scheme. As a warm-up, we first prove weak privacy. Let A be a maximal unauthorized set of parties. Let $\langle \langle s_j, v_j \rangle \rangle_{p_j \in A}$ be any vector of shares in $\{0, 1\}^q \times V$. We prove that this vector of shares has a positive probability for every secret $s \in \{0, 1\}^q$. With positive probability, the set A is chosen in Step 2 of the scheme and Procedure ChooseVertices partitions the set A to sets $C_1, \dots, C_{|A|}$, each set containing exactly one element. In this case, $\langle \langle s_j, v_j \rangle \rangle_{p_j \in A}$ are chosen with probability $1/(2^q \cdot |V|)^{|A|}$ as the shares of A . Furthermore, as G is t -dominated and $|A| \leq t$, there exists a vertex that dominates $\{v_j\}_{p_j \in A}$; i.e., the shares $\langle \langle s_j, v_j \rangle \rangle_{p_j \in A}$ can be completed to a vector of shares of all parties.

We next prove that the scheme is p -WP for the p claimed in the theorem. In the analysis, we ignore the case that the same vertex is chosen more than once in Procedure ChooseVertices (since we are choosing at most t vertices from at least 2^t vertices). Let $A \notin \Gamma$ be a maximal unauthorized set, $t' = |A|$, and $\langle \langle s_1, v_1 \rangle \rangle_{p_j \in A}$ be any possible shares of the parties in A . Let $A_1, \dots, A_\ell$ be the partition of A induced by these shares, i.e.,

Input: a secret $s \in \{0, 1\}^q$.

1. Let t be the maximum size of an unauthorized set and $G = (V, E)$ be a t -dominated graph.
2. Choose with uniform distribution a maximal unauthorized set $A \notin \Gamma$, and choose, using Procedure ChooseVertices, $|A|$ vertices $\langle v_j \rangle_{p_j \in A}$ and $|A|$ secrets $\langle s_j \rangle_{p_j \in A}$.
3. Let u be the lexicographically first vertex that dominates $\{v_j\}_{p_j \in A}$.
4. For every $p_i \notin A$ let $s_i = s$ and $v_i = u$.
5. The share of p_j is $\langle s_j, v_j \rangle$.

Reconstruction by a set $B \in \Gamma$ holding shares $\langle \langle s_j, v_j \rangle \rangle_{p_j \in B}$: Find $p_i \in B$ such that v_i dominates $\{v_j\}_{p_j \in B} \setminus \{v_i\}$ in G and output s_i .^a

Procedure ChooseVertices:

1. Let $C = A, k = 0$.
2. While $C \neq \emptyset$ do:
 - (a) Set $k = k + 1$.
 - (b) Choose with uniform distribution a subset $C_k \subseteq C$, a vertex $w_k \in V$, and a secret σ_k .
 - (c) Set $C = C \setminus C_k$ and $v_j = w_k, s_j = \sigma_k$ for every $p_j \in C_k$.

^aIf all vertices have the same vertex v , then v dominates $\{v_j\}_{p_j \in B} \setminus \{v_i\} = \emptyset$.

Figure 5: A WP secret-sharing scheme for an arbitrary access structure.

$p_j, p_{j'} \in A_d$ for some d if and only if $\text{sh}_j = \text{sh}_{j'}$. We now prove a lower bound on $\Pr \left[\Pi_A(s) = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} \right]$ for some secret $s \in \{0, 1\}^q$. Let E_A be the event that the set A is chosen in Step 2; this occurs with probability $1/M$. We only consider the case in which the E_A occurs. In this case, the shares are possible if the first set C_1 chosen in Procedure ChooseVertices is equal to A_d for some d . Furthermore, given that the first set is C_d ,

the shares of C_d and $A \setminus C_d$ are independent. That is,

$$\begin{aligned}
& \Pr \left[\Pi_A(s) = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} \right] \\
& \geq \Pr[E_A] \cdot \Pr \left[\Pi_A(s) = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} | E_A \right] \\
& = \frac{1}{M} \cdot \sum_{d=1}^{\ell} \Pr[C_1 = A_d] \Pr \left[\Pi_A(s) = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} | E_A, C_1 = A_d \right] \\
& = \frac{1}{M \cdot 2^t} \cdot \sum_{d=1}^{\ell} \Pr \left[\Pi_A(s) = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} | E_A, C_1 = A_d \right] \\
& \geq \frac{1}{M \cdot 2^t} \cdot \sum_{d=1}^{\ell} \Pr \left[\Pi_{A_d}(s) = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A_d} | E_A, C_1 = A_d \right] \\
& \quad \cdot \Pr \left[\Pi_{A \setminus A_d}(s) = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A \setminus A_d} | E_A, C_1 = A_d \right] \\
& = \frac{1}{M \cdot 2^t} \cdot \sum_{d=1}^{\ell} \frac{1}{2^q \cdot |V|} \cdot \Pr \left[\Pi_{A \setminus A_d}(s) = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A \setminus A_d} | E_A, C_1 = A_d \right].
\end{aligned} \tag{3}$$

We next prove an upper bound on $\Pr \left[\Pi_A(s') = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} \right]$ for some secret $s' \in \{0, 1\}^q$. We only consider the case $\ell > 1$ (if $\ell = 1$, we use the upper bound 1 on the probability). Let A' be the set chosen in Step 2. The shares $\langle \langle s_1, v_1 \rangle \rangle_{p_j \in A}$ can be generated only if $A = A'$ or $A' \setminus A = A_d$ for some d . That is,

$$\begin{aligned}
\Pr \left[\Pi_A(s') = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} \right] &= \Pr[E_A] \cdot \Pr \left[\Pi_A(s') = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} | E_A \right] \\
& \quad + \sum_{d=1}^{\ell} \Pr[A' \setminus A = A_d] \Pr \left[\Pi_A(s') = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} | A' \setminus A = A_d \right].
\end{aligned} \tag{4}$$

There are two cases. If the first term in the r.h.s. of (4) is bigger than the second term, then for every $s \in \{0, 1\}^q$

$$\begin{aligned}
\Pr \left[\Pi_A(s') = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} \right] &\leq 2 \Pr[E_A] \cdot \Pr \left[\Pi_A(s') = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} | E_A \right] \\
&= 2 \Pr[E_A] \cdot \Pr \left[\Pi_A(s) = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} | E_A \right] \\
&\leq 2 \cdot \Pr \left[\Pi_A(s) = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} \right],
\end{aligned}$$

where the equality follows from the fact that when the set A is chosen in Step 2, the shares of A are independent from the secret.

If the first term of the r.h.s. of (4) is smaller or equal to the second term, then

$$\begin{aligned}
\Pr \left[\Pi_A(s') = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} \right] &\leq 2 \sum_{d=1}^{\ell} \Pr[A' \setminus A = A_d] \Pr \left[\Pi_A(s') = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} | A' \setminus A = A_d \right] \\
&\leq 2 \sum_{d=1}^{\ell} \Pr \left[\Pi_{A \setminus A_d}(s') = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A \setminus A_d} | A' \setminus A = A_d \right].
\end{aligned} \tag{5}$$

We claim that

$$\begin{aligned}
& \Pr \left[\Pi_{A \setminus A_d}(s) = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A \setminus A_d} | E_A, C_1 = A_d \right] \\
&= \Pr \left[\Pi_{A \setminus A_d}(s') = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A \setminus A_d} | A' \setminus A = A_d \right]
\end{aligned} \tag{6}$$

as in the first case we iteratively choose random subsets of $A \setminus A_d$ and in the second case we iteratively choose random subsets of A' and only consider their intersection with A , i.e., we consider subsets of $A \setminus A_d$ (ignoring the subsets that do not intersect A); furthermore, in both cases the shares are independent of the secret that is shared.

To conclude, for every $s, s' \in \{0, 1\}^q$ if the first term of the r.h.s. of (4) is smaller or equal to the second term, then by (3), (6), and (5),

$$\begin{aligned}
& \Pr \left[\Pi_A(s) = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} \right] \\
& \geq \frac{1}{M \cdot 2^t \cdot 2^q \cdot |V|} \cdot \sum_{d=1}^{\ell} \Pr \left[\Pi_{A \setminus A_d}(s) = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A \setminus A_d} \mid E_A, C_1 = A_d \right] \\
& \geq \frac{1}{M \cdot 2^t \cdot 2^q \cdot |V|} \cdot \sum_{d=1}^{\ell} \Pr \left[\Pi_{A \setminus A_d}(s') = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A \setminus A_d} \mid A' \setminus A = A_d \right] \\
& \geq \frac{1}{2M \cdot 2^t \cdot 2^q \cdot |V|} \cdot \Pr \left[\Pi_A(s') = \langle \langle s_j, v_j \rangle \rangle_{p_j \in A} \right].
\end{aligned} \tag{7}$$

□

In Section 3.3.1, we provide constructions of t -dominated graphs, as summarized in the next lemma. The number of vertices in the first construction $\tilde{O}(2^t)$. Our second construction is strongly explicit, i.e., we construct a graph with $2^{O(t \log t)}$ vertices such that for every pair of vertices u, v , each one described by $O(t \log t)$ bits, one can check in time $\text{poly}(t)$ if $(u, v) \in E$.

Lemma 3.5. *For every t , there exists a t -dominated graph with $O(t^2 \cdot 2^t)$ vertices. Furthermore, for every t , there exists a strongly explicit t -dominated graph with $2^{O(t \log t)}$ vertices.*

Using Lemma 3.5 in Theorem 3.4, we obtain the following corollary.

Corollary 3.6 (WP secret sharing for general access structures). *Let Γ be an access structure in which all unauthorized sets are of size at most t and let $p = \frac{1}{O(\binom{n}{\min\{t, n/2\}} t^2 2^{2t})}$. There is a p -WP secret-sharing scheme realizing Γ with a q -bit secret and share size $O(t) + q$.*

In the special case of forbidden graphs in which $t = 2$ and binary secrets (i.e., $q = 2$), the share size is $O(1)$ and the scheme is $1/O(n^2)$ -WP. That is, the scheme achieves the same parameters as the scheme of Section 3.2, up to constant factors; however, the scheme in this subsection has perfect correctness. For an arbitrary n -party access structure, we get a $2^{-O(n)}$ -WP secret-sharing scheme with share size $O(n) + q$.

3.3.1 Constructions of t -Dominated Graphs

We next provide two constructions of t -dominated graphs, proving Lemma 3.5.

First construction. We prove that a random tournament with $N = O(t^2 \cdot 2^t)$ is t -dominated with high probability. Formally, let $N = O(t^2 \cdot 2^t)$, $V = [N]$ and for each $1 \leq i < j \leq N$ independently choose with a uniform distribution if $(i, j) \in E$ or $(j, i) \in E$. Fix a set W of size t and a vertex $u \notin V$. The probability that u dominates W is 0.5^t . Thus, the probability that there is no vertex that dominates W is

$$\left(1 - \frac{1}{2^t}\right)^{N-t} = \left(\left(1 - \frac{1}{2^t}\right)^{2^t}\right)^{O(t^2)} \leq e^{-O(t^2)}.$$

By the union bound, the probability that there is some W of size t that is not dominated by any vertex is less than

$$N^t e^{-O(t^2)} = (O(t^2 2^t))^t e^{-O(t^2)} = 2^{-(O(1)-1)t^2 + O(t \log t)} \ll 1.$$

This implies that for every t there is a t -dominated graph with $O(t^2 \cdot 2^t)$ vertices.

Second construction. The second construction is strongly explicit; however, the number of vertices is $2^{O(t \log t)}$, which is larger than the number of vertices in the first construction. The idea of the construction is that each vertex is assigned a label r (taken from a domain of size $O(t^2)$) and a set T of t labels; there is an edge from vertex $\langle r, T \rangle$ to vertex $\langle r', T' \rangle$ if $r' \in T$ and $r \notin T'$. In other words, each vertex is assigned t labels that it dominates.

Formally, we construct the following graph $G = (V, E)$, where for every $r \in [t^2 + t + 1]$ and every set $T \subset [t^2 + t + 1] \setminus \{r\}$ of size t there is a vertex $\langle r, T \rangle$ and $(\langle r, T \rangle, \langle r', T' \rangle) \in E$ if $r' \in T$ and $r \notin T'$ (in particular, G does not contain anti-parallel edges). We next show that G is t -dominated. Let $\langle r_1, T_1 \rangle, \dots, \langle r_t, T_t \rangle$ be any set of t vertices. Take T as any set of size t that contains $r_1, \dots, r_t$ (e.g., if $r_1, \dots, r_t$ are distinct, then $T = \{r_1, \dots, r_t\}$) and take r as any element in $[t^2 + t + 1] \setminus (T \cup T_1 \cup \dots \cup T_t)$ (since $|T \cup T_1 \cup \dots \cup T_t| \leq (t+1)t$, such r exists). Clearly, $\langle r, T \rangle$ dominates $\langle r_1, T_1 \rangle, \dots, \langle r_t, T_t \rangle$. The number of vertices in G is $(t^2 + t + 1) \binom{t^2 + t}{t} = 2^{O(t \log t)}$.

4 Composed Gap Access Structures

In this section, we construct *computational* secret-sharing schemes for gap access structures, as defined below. Our schemes use information-theoretic WP schemes, as those constructed in Section 3.3, and their security is based on the LPN assumption.

Next, we define a composition operation between m access structures $\Gamma_1, \dots, \Gamma_m$; for parameters $0 \leq \alpha < \beta \leq 1$, we define the composition of these access structures as a ramp scheme where the authorized sets are those that are authorized in at least β fraction of the m access structures, and the unauthorized sets are those that are authorized in less than an α fraction.

Definition 4.1. Given m fully defined access structures $\Gamma_1, \dots, \Gamma_m$ on sets of parties $P_1, \dots, P_m$ (respectively) and two real numbers $0 \leq \alpha \leq \beta \leq 1$, the composed gap access structure $\text{Gap}_{\alpha, \beta}(\Gamma_1, \dots, \Gamma_m)$ is the partially defined access structure $\Gamma = \langle \Gamma_{\text{no}}, \Gamma_{\text{yes}} \rangle$, where $B \in \Gamma_{\text{yes}}$ if and only if $|\{i : B \cap P_i \in \Gamma_i\}| \geq \beta m$ and $A \in \Gamma_{\text{no}}$ if and only if $|\{i : A \cap P_i \in \Gamma_i\}| < \alpha m$.

We will concentrate on the following two families of composed gap access structures:

- Let $P_1, \dots, P_m$ be disjoint sets of parties of size $\ell = O(\sqrt{m/\log m})$, and let $\Gamma_1, \dots, \Gamma_m$ be forbidden graph access structures with parties $P_1, \dots, P_m$, respectively; the number of parties in the composed access structure is $n = m\ell < m^{3/2}$. This family can be realized by a perfect secret-sharing scheme (even when $\alpha = \beta$), where we first share the secret using Shamir's βm -out-of- m secret-sharing scheme, producing shares $\text{sh}_1, \dots, \text{sh}_m$, and share each sh_i with the secret-sharing scheme of [39] realizing the graph access structure Γ_i . The resulting share size is $2^{\tilde{O}(\sqrt{\log n})}$. We will construct a computational secret-sharing scheme for this access structure with shares of size $O(1)$ (under the LPN assumption).
- Let $P_1, \dots, P_m$ be disjoint sets of parties of size $\ell = 0.1 \log m$, and let $\Gamma_1, \dots, \Gamma_m$ be any access structures with parties $P_1, \dots, P_m$, respectively; the number of parties in the composed access structure is $n = m\ell = O(m \log m)$. This family can be realized by a perfect secret-sharing scheme (even

when $\alpha = \beta$), where we first share the secret using Shamir's βm -out-of- m secret-sharing scheme, producing shares $\text{sh}_1, \dots, \text{sh}_m$, and share each sh_i with the secret-sharing scheme of [4] realizing the access structure Γ_i . The resulting share size is greater than $2^{0.58\ell} = n^{0.058-o(1)}$. We will construct a computational secret-sharing scheme for this access structure with shares of size $O(\log n)$ (under the LPN assumption).

4.1 An Information-Theoretic Secret-Sharing Scheme Realizing $\text{Gap}_{0.5,1}$

In Figure 6, we describe a statistical secret-sharing scheme with share size $O(1)$ realizing $\text{Gap}_{0.5,1}(\Gamma_1, \dots, \Gamma_m)$, where $\Gamma_1, \dots, \Gamma_m$ are forbidden graph access structures over disjoint sets of parties (of size $\ell = o(\sqrt{m/\log m})$). This is a simpler case as the correctness requires that a set B is authorized if $B \cap P_i$ is authorized for every $1 \leq i \leq m$.

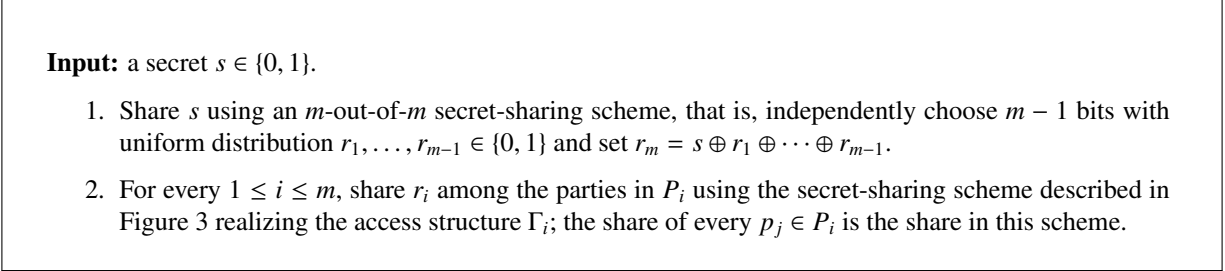


Figure 6: A statistical secret-sharing scheme for $\text{Gap}_{0.5,1}(\Gamma_1, \dots, \Gamma_m)$ for forbidden graphs access structures $\Gamma_1, \dots, \Gamma_m$.

Lemma 4.2. *Let $\Gamma_1, \dots, \Gamma_m$ be forbidden graph access structures over disjoint sets of parties of size $\ell = o(\sqrt{m/\log m})$. The scheme described in Figure 6 is a statistical secret-sharing scheme with share size $O(1)$ realizing $\text{Gap}_{0.5,1}(\Gamma_1, \dots, \Gamma_m)$.*

Proof. We first explain the correctness of the secret-sharing scheme. Let B be an authorized set of $\text{Gap}_{0.5,1}(\Gamma_1, \dots, \Gamma_m)$, thus, $B \cap P_i$ is an authorized set of Γ_i , for every $1 \leq i \leq m$. By the perfect correctness of the secret-sharing scheme described in Figure 3, the parties in $B \cap P_i \subset B$ can reconstruct r_i ; thus, the parties in B can reconstruct all m shares of the m -out-of- m secret-sharing scheme, i.e., they can reconstruct s .

We next prove the statistical security of the secret-sharing scheme. Let A be a maximal unauthorized set of $\text{Gap}_{0.5,1}(\Gamma_1, \dots, \Gamma_m)$. Since A is unauthorized, there are at least $m/2$ indices i such that $A \cap P_i$ is unauthorized in Γ_i ; i.e. it is a non-edge of the corresponding graph; for each such i , the two parties in $A \cap P_i$ get the shares $\langle 2, 2 \rangle$ with probability at least $1/\ell^2$. If for at least one i , the parties in $A \cap P_i$ get the shares $\langle 2, 2 \rangle$, then the parties in A get no information on at least one share in the m -out-of- m secret-sharing scheme; hence, get no information on the secret. Thus, the probability that the parties in A get some information on the secret is at most

$$\left(1 - O\left(\frac{1}{\ell^2}\right)\right)^{0.5m} = \left((1 - O(1/\ell^2))^{O(\ell^2)}\right)^{-\omega(\log m)} < e^{-\omega(\log m)}.$$

□

4.2 A Computational Secret-Sharing Scheme $\text{Gap}_{\alpha,\beta}$

In Figure 7, we describe a computational secret-sharing scheme realizing $\text{Gap}_{\alpha,\beta}(\Gamma_1, \dots, \Gamma_m)$ for constants α, β such that $\beta > 0.5$ and $\alpha \leq 1 - h(\beta)$. The security of our scheme holds under the LPN assumption, i.e., the intractability assumption $\text{LPN}(c'k, \gamma' / \sqrt{k})$ for some constants $c', \gamma' > 0$. We consider a family of access structures, where for every n an n -party access structure in this family is defined by composing access structures $\Gamma_1, \dots, \Gamma_m$ such that each Γ_i is an access structure with a disjoint set of parties that can be realized by a p -WP secret-sharing scheme Π_i for $p = 1/O(\sqrt{m})$. We will use Lemma 2.8 to simulate this scheme. That is, we would like to share a bit z_i ; however, we are only given a noisy version $y_i = \frac{p}{1+p}\text{-BSC}(z_i)$, that is $\Pr[y_i = z_i] = 1/(p+1)$ and $\Pr[y_i = 1 - z_i] = p/(p+1)$. By Lemma 2.8, there is a function $g_i(y_i)$ generating the shares of $A \cap P_i$ for a secret z_i from its noisy version y_i . We require that the sharing of the secret and the reconstructing of the secret from shares of any authorized set can be done in $\text{poly}(m)$ time. Furthermore, to be able to compute $g(y_i)$ efficiently, we require that for every secret $s \in \{0, 1\}$, every set $A \notin \Gamma_i$, and every possible shares $\langle \text{sh}_j \rangle_{p_j \in A}$, the probability $\Pr[\Pi_A(s) = \langle \text{sh}_j \rangle_{p_j \in A}]$ can be computed in $\text{poly}(m)$ time. We refer to such a scheme as a simulatable p -WP secret-sharing scheme.

Input: a secret $s \in \{0, 1\}$.

1. Let $m = m(n)$, c be a constant such that $1/(1 - h(\beta)) < c < 1/\alpha$, and $k = (m+1)/c$.
2. Share s using a random binary code; that is, choose a random matrix $C \leftarrow \mathbb{F}_2^{(m+1) \times k}$ and a random $x \leftarrow \mathbb{F}_2^k$ such that the zero coordinate of Cx is s and let $\text{sh}_1, \dots, \text{sh}_m$ be the last m coordinates of Cx .
3. For every $1 \leq i \leq m$, share sh_i among the parties in P_i using a $1/O(\sqrt{m})$ -WP secret-sharing scheme realizing the access structure Γ_i ; the share of each $p_j \in P_i$ is its share in this scheme. The public information sh_0 is C .

Figure 7: A computational secret-sharing scheme for $\text{Gap}_{\alpha,\beta}(\Gamma_1, \dots, \Gamma_m)$ for access structures $\Gamma_1, \dots, \Gamma_m$, where Γ_i is an access structure with disjoint set of parties P_i that can be realized by a $1/O(\sqrt{m})$ -simulatable WP secret-sharing scheme, assuming the hardness of LPN.

Theorem 4.3 (WP-based composed gap secret sharing). *Let α, β be constants such that $\beta > 1/2$ and $0 \leq \alpha \leq 1 - h(\beta)$, c be a constant such that $1/(1 - h(\beta)) < c < 1/\alpha$, and $p : \mathbb{N} \rightarrow [0, 1]$ be a function such that $p(m) \geq \gamma/\sqrt{m}$ for some constant $\gamma > 1$. There exist constants $c', \gamma' > 0$ such that assuming that $\text{LPN}(c'k, \gamma' / \sqrt{k})$ holds, there is a computational secret-sharing scheme with share size $s(m)$ realizing the family of composed access structures $\text{Gap}_{\alpha,\beta}(\Gamma_1, \dots, \Gamma_m)$, where every Γ_i is an access structure that can be realized by a simulatable $p(m)$ -WP secret-sharing scheme with share size $s(m)$.*

Proof. The secret-sharing scheme guaranteed by the theorem is described in Figure 7. We first explain its correctness. Let $B \in \text{Gap}_{\alpha,\beta}(\Gamma_1, \dots, \Gamma_m)$, thus, $B \cap P_i \in \Gamma_i$ for at least βm indices. By the perfect correctness of a WP secret-sharing scheme, the parties in $B \cap P_i \subset B$ can reconstruct sh_i for each such Γ_i ; thus, the parties in B can reconstruct at least βm shares in the secret-sharing scheme of the random code C . By Theorem 2.10, the distance of the code C is at least $(1 - \beta - 2/(m+1))(m+1) = (1 - \beta)(m+1) + 2$ except with probability $2^{k - (m+1)(1 - h(1 - \beta + 2/(m+1)))} = 2^{k(1 - c(1 - h(\beta - 2/(m+1))))}$. For large enough m , this probability is at most $2^{-\Theta(k)}$ (since c and β are constants such that $c > 1/(1 - h(\beta))$). By Theorem 2.9, the secret can be recovered from any $m - ((1 - \beta)(m+1) + 2) + 2 < \beta m$ shares. That is, except for a negligible probability, the parties in B can recover the secret.

We next argue the computational security of the scheme. By choosing appropriate constants $c', \gamma' > 0$ and using Lemma 2.12 and the fact that $c\alpha < 1$, the $\text{LPN}(c'k, \gamma'/\sqrt{k})$ implies the $\text{LPN}(ck, \gamma/\sqrt{k}, \alpha)$ assumption. Furthermore, by Remark 2.13, the αm exposed bits can be chosen by an adversary (prior to seeing C). Let $\mathcal{A}$ be a poly-time adversary trying to break the security of the scheme, that is, for a security parameter n , the adversary $\mathcal{A}$ chooses integers m, ℓ such that $n = m\ell$, and m access structures $\Gamma_1, \dots, \Gamma_m$, each Γ_i is an access structure with ℓ disjoint parties that can be realized by a simulatable $p(m)$ -WP secret-sharing scheme. Since the sharing for each Γ_i can be computed in time $\text{poly}(m)$, then $\ell \leq \text{poly}(m)$ and $m \leq n \leq m \cdot \text{poly}(m)$. The adversary also chooses an unauthorized set A of $\text{Gap}_{\alpha, \beta}(\Gamma_1, \dots, \Gamma_m)$. The adversary sends $\Gamma_1, \dots, \Gamma_m$ and A to the Challenger, gets a code C and the shares of A in the scheme for a random secret s , and outputs a secret s' such that $s = s'$ with probability $1/2 + \varepsilon(n)$ for some function $\varepsilon(n)$. We show that we can use $\mathcal{A}$ to construct a poly-time adversary $\mathcal{B}$ that breaks the $\text{LPN}(ck, \gamma/\sqrt{k}, \alpha)$ assumption with advantage $\varepsilon(n)$. By the $\text{LPN}(ck, \gamma/\sqrt{k}, \alpha)$ assumption, the advantage $\varepsilon(n)$ is a negligible function of k , hence a negligible function of n .

Let Γ_i be an access structure realized by a simulatable $p(m)$ -WP secret-sharing scheme; since $p(m) \geq \gamma/\sqrt{k}$, this scheme is a $\gamma/\sqrt{k}$ -simulatable $p(m)$ -WP secret-sharing scheme. Furthermore, let A be an unauthorized in Γ_i and

$$\delta = \frac{\gamma/\sqrt{k}}{1 + \gamma/\sqrt{k}} \geq \frac{\gamma}{2\sqrt{k}}.$$

Let $z_i \in \{0, 1\}$ and $y_i = \delta\text{-BSC}(z_i)$. By Lemma 2.8, there is a function $g_i(z_i)$ that generates shares of A for a secret z_i from the bit y_i ; the function g_i can be computed in $\text{poly}(m) = \text{poly}(k)$ time.

We next describe the adversary $\mathcal{B}$ interacting with an encoder:

- **Input:** The security parameter 1^k .
- Invoke $\mathcal{A}$ with input 1^k , get $\Gamma_1, \dots, \Gamma_m$ and a set A that is an unauthorized set in $\text{Gap}_{\alpha, \beta}(\Gamma_1, \dots, \Gamma_m)$.
- Let $I = \{i \in [m] : A \cap P_i \in \Gamma_i\}$; since A is unauthorized, $|I| \leq \alpha m$.
- Send $I \cup \{0\}$ to an encoder, get as a challenge a code $C \in \mathbb{F}_2^{(m+1) \times k}$ and a word $y \in \{0, 1\}^{m+1}$ such that y is either uniformly distributed or computed as $z = Cx, y = z + e$ for a uniformly distributed x , $e_i = 0$ for every $i \in I \cup \{0\}$, and each other e_i is sampled from a Bernoulli distribution with parameter δ .
- Prepare shares for the parties in A as follows:
 1. For every $i \in I$, share y_i among the parties in P_i using the simulatable $1/O(\sqrt{m})$ -WP secret-sharing scheme realizing Γ_i ; the share of every $p_j \in A \cap P_i$ is the share in this scheme.
 2. For every $i \in [m] \setminus I$, prepare shares for the parties in $A \cap P_i$ as $g_i(y_i)$.
- Give to $\mathcal{A}$ the matrix C and the shares it prepared for the parties in A . In response, it gets a secret s' .
- If $s' = y_0$, then output y is a noisy codeword, otherwise output y is random.

If y was generated as a random word, then y_0 is independent of the rest of the coordinates of y ; thus, in this case $s' = y_0$ with probability exactly $1/2$. On the other hand, if y was generated as a noisy version of a random code-word (i.e., $z = Cx; y = z + e$, where $e_i = 0$ for $i \in I \cup \{0\}$ and each e_i for $i \in [m] \setminus I$ is independently sampled from a Bernoulli distribution with parameter δ), then the shares of A that $\mathcal{B}$ generated are distributed as if the dealer shares z in Step 3, that is, as in the security game for the scheme of Figure 7, thus, $s = y_0$ with probability $1/2 + \varepsilon(n)$; i.e., $\mathcal{B}$ distinguishes between the two distributions with advantage $\varepsilon(n)$. $\square$

We next consider the two families of composed access structures discussed in the beginning of Section 4.

Corollary 4.4. *Let $\beta > 1/2$ and $\alpha < 1 - h(\beta)$. There are constants $c', \gamma' > 0$ such that assuming that $\text{LPN}(c'k, \gamma' / \sqrt{k})$ holds, there is a computational secret-sharing scheme with share size $O(1)$ for the family of access structures $\text{Gap}_{\alpha, \beta}(\Gamma_1, \dots, \Gamma_m)$, where each Γ_i is a forbidden graph access structure with $\ell(m) = m^{1/4}$ disjoint parties.*

Proof. We consider the secret-sharing scheme described in Figure 5 to realize each Γ_i . This scheme is $1/O(\ell(m)^2)$ -WP, where $1/(\ell(m))^2 = 1/\sqrt{m}$. Furthermore, for each unauthorized set (of size 2), the probability of each pair of shares can be computed in time $O((\ell(m))^2)$ by looping over all unauthorized sets A of size 2 and summing the conditional probability of these shares given that the unauthorized set A was chosen by the scheme. $\square$

Corollary 4.5. *Let $\beta > 1/2$ and $\alpha < 1 - h(\beta)$. There are constants $c', \gamma' > 0$ such that assuming that $\text{LPN}(c'k, \gamma' / \sqrt{k})$ holds, there is a computational secret-sharing scheme with share size $\log m$ for the family of access structures $\text{Gap}_{\alpha, \beta}(\Gamma_1, \dots, \Gamma_m)$, where each Γ_i is an access structure with $\ell(m) = 0.1 \log m$.*

Proof. We consider the secret-sharing scheme described in Figure 5 to realize each Γ_i , where the maximum size of an unauthorized set in Γ_i is at most $\ell = \ell(m)$. This scheme is $1/\tilde{O}(2^{3\ell})$ -WP, where $1/\tilde{O}(1/2^{3\ell}) > 1/\sqrt{m}$. For each unauthorized set (of size at most $0.1 \log m$), the probability of each tuple of shares can be computed in time $2^{O(\log m)} = \text{poly}(m)$ by looping over all unauthorized sets A and summing the conditional probability of these shares given that the unauthorized set A was chosen by the scheme. Furthermore, in this case the $0.1 \log m$ -dominated graph of size $\tilde{O}(m^{0.1})$ can be efficiently sampled by the dealer. $\square$

Remark 4.6. For composition of forbidden graph access structures, there is a variant of the above scheme in which the shares are a bit. The scheme is obtained from the scheme described in Figure 7 by using the simpler scheme described in Figure 4 as the WP scheme in Step 3 (instead of the scheme described in Figure 5). This results in a scheme with 1-bit shares. As the scheme described in Figure 4 is $1/O(\ell^2)$ -WP, the resulting scheme is secure under the intractability of $\text{LPN}(c'k, \gamma' / \sqrt{k})$.

However, the scheme described in Figure 4 is not perfectly correct – with probability at most $2/\ell$ it can return $\perp$. We use the fact that the scheme never errs; rather, with small probability it outputs $\perp$. We will show that for $\beta' = \Omega(m^{-0.75})$ any authorized set $B \in \text{Gap}_{\alpha, \beta + \beta'}(\Gamma_1, \dots, \Gamma_m)$ can reconstruct the secret with high probability. For such B , we have $B \cap P_i \in \Gamma_i$ for at least $\beta m + \Omega(m^{0.25})$ indices. As before, assume that $m \geq n^{0.8}$, thus, $\ell \leq m^{0.25}$. By the Chernoff inequality, the probability that in more than $m \cdot 4/\ell = O(m^{0.25})$ coordinates, the reconstruction of sh_i will return $\perp$ is at most $e^{-O(m^{0.25})}$. Thus, with high probability, the authorized set B can reconstruct at least βm shares of the ramp scheme and reconstruct the secret.

We do not know how to generalize these ideas for composition of arbitrary access structures, as we do not know how to generalize the scheme described in Figure 4 to arbitrary access structures while having small reconstruction error.

5 Weakly Private Decomposable Randomized Encoding and MPC protocols

In this section we construct WP decomposable randomized encoding (DRE) (see Section 5.1) and WP MPC protocols (see Section 5.2).

5.1 A WP Decomposable Randomized Encoding (DRE)

We next show that every function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ has a WP DRE of size $\tilde{O}(n^2)$ and prove that there are explicit functions f such that every WP DRE for f is of size $\Omega(n^2 / \log n)$. As mentioned in the introduction, the notion of DRE is closely related to notions such as garbling and n -party PSM protocols.

We start with our positive result, where we construct a very efficient WP DRE for any $f : \{0, 1\}^n \rightarrow \{0, 1\}$, by constructing a DRE for a related function g . The idea is that the function g is, on one hand, very simple and hence has a very efficient perfect DRE (independently of the complexity of f) and, on the other hand, g 's output allows computing f with some (weak) level of privacy.

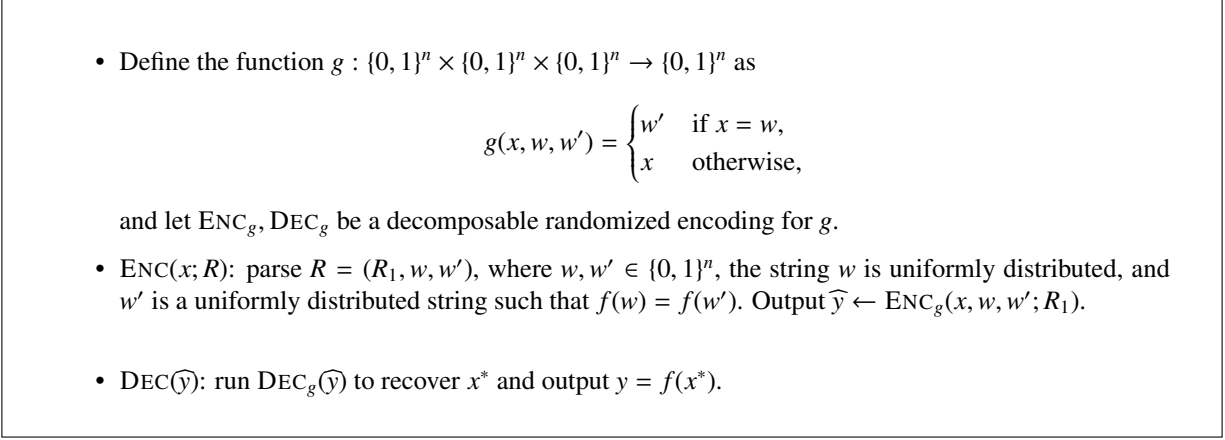


Figure 8: A WP decomposable randomized encoding for any function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ with $O(n^2 \cdot \log^2 n \cdot \log \log n)$ size.

Theorem 5.1. *Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be a function. The randomized encoding scheme for f , described in Figure 8, is a p -WP DRE for $p = 2^{-O(n)}$, and has size $\ell \leq O(n^2 \cdot \log^2 n \cdot \log \log n)$.*

Proof. The correctness of the construction follows directly from those of $\text{ENC}_g, \text{DEC}_g$ and the fact that $f(w) = f(w')$. The function g can be implemented by n Boolean formulae

$$g_i(x, w, w') := \left(w'_i \wedge \bigwedge_i (x_i = y_i) \right) \vee \left(x_i \wedge \bigvee_i (x_i \neq y_i) \right);$$

each g_i is computed by a formula of depth $\log n + O(1)$. By Lemma 2.15, there is a perfect DRE for each g_i of size $O(n^2 \log^2 n \log \log n)$; thus, the DRE for g is as claimed..

We now argue p -WP for $p = 2^{-O(n)}$. Consider any inputs $x_1, x_2 \in \{0, 1\}^n$ such that $f(x_1) = f(x_2) = y$ for some y , and any encoding $\widehat{y} \in \{0, 1\}^\ell$. Let $x^* = \text{DEC}_g(\widehat{y})$. As shorthands, we call $\Pr[\text{ENC}(x_1) = \widehat{y}]$ and $\Pr[\text{ENC}(x_2) = \widehat{y}]$ the LHS and RHS probabilities, respectively.

We now analyze the LHS and RHS probabilities. Two easy cases are when (1) $x_1 = x_2$, or (2) $f(x^*) \neq y$. p -WP is trivially satisfied in (1) as the LHS and RHS probabilities are the same, and in (2) as the two probabilities equal 0, by correctness. For the general case, let W_1, W_2 denote the sets of possible rejection sampling results (w, w') during ENC such that $g(x_1, w, w') = x^*$ and $g(x_2, w, w') = x^*$, respectively. The security of the underlying (standard) decomposable randomized encoding guarantees that

$$\forall (w_1, w'_1) \in W_1, (w_2, w'_2) \in W_2, \quad \Pr[\text{ENC}_g(x_1, w_1, w'_1)] = \Pr[\text{ENC}_g(x_2, w_2, w'_2)] = \varepsilon.$$

The LHS and RHS probabilities are given by

$$LHS = \sum_{(w,w') \in W_1} \Pr[w, w' \text{ sampled during ENC}] \cdot \varepsilon, \quad RHS = \sum_{(w,w') \in W_2} \Pr[w, w' \text{ sampled during ENC}] \cdot \varepsilon. \quad (8)$$

Consider the three possible cases:

- $x_1 \neq x^*, x_2 \neq x^*$. In this case, W_1 and W_2 each only contain a single pair: $(w_1 = x_1, w'_1 = x^*)$, and $(w_2 = x_2, w'_2 = x^*)$. We have $LHS = RHS$, and p -WP holds.
- $x_1 = x^*, x_2 \neq x^*$. In this case, W_1 contains all possible pairs (w_1, w'_1) , as long as $w_1 \neq x_1$, and W_2 contains a single pair $(w_2 = x_2, w'_2 = x^*)$. In particular, $W_2 \subseteq W_1$. Hence $LHS \geq RHS$, and p -WP holds.
- $x_1 \neq x^*, x_2 = x^*$. This case is analogous to the previous, with W_1 contains a single pair, and W_2 contains many. We loosely bound $RHS \leq 1 \cdot \varepsilon$, and $LHS \geq 1/(2^n)^2 \cdot \varepsilon$. Hence $LHS \geq 2^{-2n} RHS$, i.e. p -WP holds for $p = 2^{-O(n)}$.

□

5.1.1 An $\Omega(n^2 / \log n)$ Size Lower Bound

Our next goal is to show that the complexity of the above WP DRE scheme is essentially optimal for some functions f . We start with some notations.

Function restrictions. For a subset $S \subseteq [n]$ with complement $\bar{S}$ and a partial input $z \in \{0, 1\}^{\bar{S}}$, let $f_{S|z}$ denote the restriction of f on S via z :

$$f_{S|z}(x) := f(x'), \text{ where } x \in \{0, 1\}^S, \text{ and } x'^{(i)} = \begin{cases} x^{(i)} & \text{if } i \in S \\ z^{(i)} & \text{otherwise.} \end{cases}$$

Let $g_S(f)$ count the (log of) number of distinct, non-constant restrictions of f on S :

$$g_S(f) := \log(\#\{f_{S|z} : z \in \{0, 1\}^{\bar{S}}, f_{S|z} \text{ not constant}\}).$$

Given a DRE scheme, let ENC_S denote the output bits that are determined by the input bits at positions in S . Let $\ell_S \leq \ell$ denote the number of such output bits. For example, for a partial input $z \in \{0, 1\}^{\bar{S}}$ and randomness R , we write $\widehat{y}_{\bar{S}} = \text{ENC}_{\bar{S}}(z; R)$ and say $\widehat{y}_{\bar{S}}$ is a partial encoding with respect to z . The bit-length of the partial encoding is denoted by $\ell_{\bar{S}} = |\widehat{y}_{\bar{S}}|$.

We first show in the following claim that WP privacy of a DRE scheme requires a partial encoding $\widehat{y}_{\bar{S}}$ with respect to some partial input $z_1 \in \{0, 1\}^{\bar{S}}$ to be explainable by any other partial input z_2 , as long as $f_{S|z_1}$ and $f_{S|z_2}$ are not constant.

Clm 5.2 (WP of partial inputs). *Let $\mathcal{R}, \text{ENC}, \text{DEC}$ be a WP decomposable randomized encoding for some function $f : \{0, 1\}^n \rightarrow \{0, 1\}$. For any $S \subseteq [n]$, partial inputs $z_1, z_2 \in \{0, 1\}^{\bar{S}}$ such that the restrictions $f_{S|z_1}, f_{S|z_2}$ are not constant, and any partial encoding $\widehat{y}_{\bar{S}} \in \{0, 1\}^{\ell_{\bar{S}}}$,*

$$\Pr_{R \leftarrow \mathcal{R}} [\text{ENC}_{\bar{S}}(z_1; R) = \widehat{y}_{\bar{S}}] > 0 \implies \Pr_{R \leftarrow \mathcal{R}} [\text{ENC}_{\bar{S}}(z_2; R) = \widehat{y}_{\bar{S}}] > 0.$$

Proof. Since $f_{S|z_1}$ is non-constant, in particular not constantly zero, there exists an extension $x_1 \in \{0, 1\}^n$ to the partial input $z_1 \in \{0, 1\}^{\bar{S}}$ such that $f(x_1) = 1$. Similarly, there also exists an extension x_2 to the partial input z_2 such that $f(x_2) = 1$.

$$f(x_1) = f(x_2), \quad \text{and for } i \in \bar{S}, \quad x_1^{(i)} = z_1^{(i)}, \quad x_2^{(i)} = z_2^{(i)}.$$

By assumption, there exists $R_1 \in \mathcal{R}$ such that $\text{ENC}_{\bar{S}}(z_1; R_1) = \widehat{y}_{\bar{S}}$. Let $\widehat{y} = \text{ENC}(x_1; R_1)$. It suffices to show that $\Pr_{R_2 \leftarrow \mathcal{R}}[\text{ENC}(x_2; R_2) = \widehat{y}] > 0$, as the event $\text{ENC}(x_2; R_2) = \widehat{y}$ implies $\text{ENC}_{\bar{S}}(z_2; R_2) = \widehat{y}_{\bar{S}}$. The desired inequality follows from WP (for any $p > 0$):

$$\Pr_{R_2 \leftarrow \mathcal{R}}[\text{ENC}(x_2; R_2) = \widehat{y}] \geq p \cdot \Pr_{R_1 \leftarrow \mathcal{R}}[\text{ENC}(x_1; R_1) = \widehat{y}] > 0.$$

□

Building onto the previous claim, we next show that the bit-length of a partial encoding ℓ_S is at least $g_S(f)/2$, where $g_S(f)$ counts the log of the number of non-constant restrictions to S .

Clm 5.3 (Size lower-bound for partial encodings). *Let $\mathcal{R}, \text{ENC}, \text{DEC}$ be a WP decomposable randomized encoding for some function $f : \{0, 1\}^n \rightarrow \{0, 1\}$. For any $S \subseteq [n]$, we have $2\ell_S \geq g_S(f)$.*

Proof. Consider any $S \subseteq [n]$. If there are no non-constant restrictions on S , then we have $g_S(f) = 0$ and we are done. Otherwise, let $\widehat{y}_{\bar{S}}$ be a partial encoding that can explain all partial inputs $z \in \{0, 1\}^{\bar{S}}$ such that $f_{S|z}$ are non-constant:

$$\forall z \in \{0, 1\}^{\bar{S}} \text{ s.t. } f_{S|z} \text{ not constant}, \quad \exists R \text{ s.t. } \text{ENC}_{\bar{S}}(z, R) = \widehat{y}_{\bar{S}}. \quad (9)$$

(Such a partial encoding can be obtained by choosing an arbitrary partial input z_1 such that $f_{S|z_1}$ is non-constant, an arbitrary randomness R_1 , and computing $\widehat{y}_{\bar{S}} = \text{ENC}_{\bar{S}}(z_1; R_1)$. By Claim 5.2, $\widehat{y}_{\bar{S}}$ also explains all other partial input z_2 such that $f_{S|z_2}$ is non-constant.)

For every such z, R that explains $\widehat{y}_{\bar{S}}$ (as in (9)), we examine the possible extensions to $\widehat{y}_{\bar{S}}$ into a complete encoding:

$$\text{Ext}_{S,R} := \{\text{ENC}_{[i]}(b, R) : i \in S, b \in \{0, 1\}\}. \quad (10)$$

By decomposability of the DRE scheme, the extension to $\widehat{y}_{\bar{S}}$ from a partial encoding w.r.t. $z \in \{0, 1\}^{\bar{S}}$ into a complete encoding w.r.t. $x \in \{0, 1\}^n$ is exactly obtained by choosing $\text{ENC}_{[i]}(x^{(i)}, R)$ from the materials $\text{Ext}_{S,R}$ computed above in (10). Then by the correctness of the DRE scheme, this extended encoding w.r.t. x can be decoded by DEC to recover $f(x)$. Therefore, the materials $\text{Ext}_{S,R}$, together with $\widehat{y}_{\bar{S}}$, determine the truth table of $f_{S|z}$.

In summary, we showed that for every partial input z one can compute a set of materials $\text{Ext}_{S,R}$ that determine the truth table of $f_{S|z}$, as long as $f_{S|z}$ is non-constant. This implies the bit-length of $\text{Ext}_{S,R}$ to be at least $g_S(f)$. Therefore, we can conclude

$$|\text{Ext}_{S,R}| = 2\ell_S \geq g_S(f).$$

□

Finally, we rely on known results taken from [6] to conclude a $\Omega(n^2 / \log n)$ lower bound on the size of WP DRE schemes.

Fact 5.4. Let V denote any partition of $[n]$. There exist explicit functions $f : \{0, 1\}^n \rightarrow \{0, 1\}$ such that

$$\max_V \sum_{S \in V} g_S(f) \geq \Omega(n^2 / \log n).$$

Theorem 5.5. There exist explicit functions $f : \{0, 1\}^n \rightarrow \{0, 1\}$ such that any WP-decomposable randomized encoding for f has size $\ell \geq \Omega(n^2 / \log n)$.

Proof. Note that for any partition V of $[n]$, we have $\ell = \sum_{S \in V} \ell_S$. The lower bound follows from Claim 5.3 and Fact 5.4. $\square$

5.2 WP MPC Protocols

We next present WP MPC protocols with communication complexity $O(nk)$ secure against a semi-honest adversary. We construct protocols in two models: (1) in the plain model assuming honest majority, and (2) in the OT-hybrid model, where there is no bound on the number of semi-honest parties. Both protocols have the same structure, similar to our DRE construction above. That is, given a function $f : (\{0, 1\}^n)^k \rightarrow \{0, 1\}$, which k parties want to compute, they instead compute a related function g , which is much simpler to compute than f , but leaks some information. This function g is computed by a perfect MPC protocol.

Formally, let $1 \leq t \leq k$ be an upper bound on the number of semi-honest parties. For simplicity of the notations, we assume that the k parties in the protocol are parties $0, \dots, k-1$. Define the function

$$g_{k,t} : \left((\{0, 1\}^n)^{2k+1} \times \{0, \dots, t\} \right)^{t+1} \times (\{0, 1\}^n)^{k-t-1} \rightarrow (\{0, 1\}^n)^k,$$

where the input of the j -th party for $0 \leq j \leq t$ is $x_j, (x_\ell^j, y_\ell^j)_{0 \leq \ell \leq k-1}, r_j$, the input of the j -th party for $t \leq j \leq k-1$ is x_j , and the value of $g_{k,t}$ on these inputs is computed as follows: Let $i = \sum_{0 \leq j \leq t-1} r_j \bmod (t+1)$;

$$\begin{aligned} \text{If } f(x_0^i, \dots, x_{k-1}^i) = f(x_0, \dots, x_{k-1}) \text{ and } x_j^i = x_j \text{ for } 0 \leq j \leq k-1 \text{ then output } y_0^i, \dots, y_{k-1}^i, \\ \text{else output } x_0, \dots, x_{k-1}. \end{aligned} \quad (11)$$

Protocol $\Pi_f(x_0, \dots, x_{k-1})$: Let $\Pi_{g_{k,t}}$ be a perfect t -private protocol for $g_{k,t}$.

1. The j -th party, for $0 \leq j \leq t-1$, samples with uniform distribution $(x_\ell^j, y_\ell^j)_{0 \leq \ell \leq k-1} \in (\{0, 1\}^n)^{2k}$ and $r_j \in \{0, \dots, t\}$.
2. Secularly compute

$$g\left((x_0, (x_\ell^0, y_\ell^0)_{0 \leq \ell \leq k-1}, r_0), \dots, (x_t, (x_\ell^t, y_\ell^t)_{0 \leq \ell \leq k-1}, r_t), x_{t+1}, \dots, x_{k-1}\right)$$

using $\Pi_{g_{k,t}}$; get output $z_0, \dots, z_{k-1}$.

3. Output $f(z_0, \dots, z_{k-1})$.

Figure 9: A WP k -party protocol for a function $f : (\{0, 1\}^n)^k \rightarrow \{0, 1\}$, in the $g_{k,t}$ -hybrid model.

Lemma 5.6. If $\Pi_{g_{k,t}}$ is a perfect t -private protocol for $g_{k,t}$, then Protocol Π_f described in Figure 9 is a $\frac{1}{(t+1)2^{2kt}}$ -WP MPC protocol for f .

Proof. Correctness is trivial since $f(z_0, \dots, z_{k-1}) = f(z_0, \dots, x_{k-1})$, by the definition of $g_{k,t}$. We argue the privacy in the $g_{k,t}$ -hybrid model (i.e., assuming that g is computed by a trusted party). Let $A \subset \{0, \dots, k-1\}$ be a set of semi-honest parties of size at most t . Let $x_0, \dots, x_{k-1}, x'_0, \dots, x'_{k-1}$ such that $f(x_0, \dots, x_{k-1}) = f(x'_0, \dots, x'_{k-1})$ and $x_j = x'_j$ for every $j \in A$. Fix a view of the parties in A (for both inputs)

$$(x_j)_{j \in A}, ((x_\ell^j, y_\ell^j)_{0 \leq \ell \leq k-1}, r_j)_{j \in A \cap \{0, \dots, t\}}, z_0, \dots, z_{k-1}$$

The interesting case is when $z_0, \dots, z_{k-1} = x_0, \dots, x_{k-1}$. The probability of this view given $x_0, \dots, x_{k-1}$ is at most the probability that the parties in A choose the randomness in the view, i.e., $1/((t+1)2^{2nk})^{|A \cap \{0, \dots, t\}|}$. We next prove a lower bound on the probability of this view given $x'_0, \dots, x'_{k-1}$. Fix a party $i \in \{0, \dots, t\} \setminus A$. In this case, $i = \sum_{j=0}^t r_j$ with probability $1/(t+1)$ and $x_j^i = x'_j$ and $y_j^i = x_j$ for every $0 \leq j \leq k-1$ with probability 2^{2nk} . Thus, the probability of this view given $x'_0, \dots, x'_{k-1}$ is at least

$$\left(\frac{1}{(t+1)2^{2nk}} \right)^{|A \cap \{0, \dots, t\}|} \cdot \frac{1}{(t+1)2^{2nk}}.$$

This implies that the protocol is $\frac{1}{(t+1)2^{2nk}}$ -WP. $\square$

To construct an MPC protocol for f , we need to explain which MPC protocol $\Pi_{g_{k,t}}$ we use. Note that, for each $0 \leq j \leq t$, the equality $f(x_0^j, \dots, x_{k-1}^j) = f(y_0^j, \dots, y_{k-1}^j)$ can be locally computed without any communication. Using this information, the function $g_{k,t}$ can be computed by a Boolean or Arithmetic circuit with fan-in 2 and size $\text{poly}(k) \cdot n$.

If we assume an honest majority, we can use the BGW protocol [14] to compute $g_{k, \lceil (k+1)/2 \rceil}$. The communication complexity of the BGW protocol is polynomial in the number of parties and linear in the size of the arithmetic circuit; this results in a protocol with communication complexity $\text{poly } k \cdot n$ for f .

Without any assumptions on the number of semi-honest parties, we can use the GMW protocol [30] to compute $g_{k,k-1}$ in the OT hybrid model. The number of OT calls in the GMW protocol is polynomial in the number of parties and linear in the size of the Boolean circuit; this results in a protocol for f with $\text{poly } k \cdot n$ OT calls.

Thus, we obtain the following:

Theorem 5.7. *Let $f : (\{0, 1\}^n)^k \rightarrow \{0, 1\}$ be a function. Then,*

- *There is a 2^{2kn} -WP secure k -party protocol for f assuming an honest majority with communication complexity $\text{poly } k \cdot n$.*
- *There is a 2^{2kn} -WP secure k -party protocol for f in the OT-hybrid model in which the number of calls to OT is $\text{poly } k \cdot n$.*

Remark: Similar protocols can be constructed for general k -private, k -party MPC with correlated randomness, by using k -out-of- k secret sharing of the values. In this case, the input $\mathbf{x} = (x^1, \dots, x^k)$ is of total length $k \cdot n$, and sharing two such inputs $\mathbf{x}_1$ and $\mathbf{x}_2$ among the k parties will lead to $O(k^2 n)$ communication and correlated-random complexity.

5.3 WP MPC in the Correlated-Randomness model

The idea of our protocol, in Figure 10 below, is similar to the idea behind our DRE construction above. That is, given a function $f : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}$, that we want to compute, we instead compute a related function g , which is much simpler to compute than f , but leaks some information. Informally, we choose two random inputs (x_1, y_1) and (x_2, y_2) for f (using the correlated randomness) such that the values of f on both are the same. If the actual input (x, y) equals (x_1, y_1) we output (x_2, y_2) . Otherwise, we output (x, y) . The extra twist here is that we do not want Alice and Bob to know these two random inputs, as this allows them to compare the final output against (x_2, y_2) and know for sure which case has happened. This clearly breaks the deniability of the final output, and hence the WP property of the protocol. So instead, the correlated randomness additively shares these values among them.

Let Π_g be a two-party protocol in the correlated-randomness model for the function $g : \{0, 1\}^{5n} \times \{0, 1\}^{5n} \rightarrow \{0, 1\}^{2n}$.^a

$$g((x, x_{1,1}, y_{1,1}, x_{2,1}, y_{2,1}), (y, x_{1,2}, y_{1,2}, x_{2,2}, y_{2,2})) = \begin{cases} (x_{2,1} + x_{2,2}, y_{2,1} + y_{2,2}) & \text{if } x = x_{1,1} + x_{1,2} \\ & \& y = y_{1,1} + y_{1,2} \\ (x, y) & \text{otherwise.} \end{cases}$$

Correlated Randomness: sample x_1, y_1, x_2, y_2 from $\{0, 1\}^n$ such that $f(x_1, y_1) = f(x_2, y_2)$, via rejection sampling. Additively share each of these 4 values as $x_1 = x_{1,1} + x_{1,2}$, $y_1 = y_{1,1} + y_{1,2}$, $x_2 = x_{2,1} + x_{2,2}$ and $y_2 = y_{2,1} + y_{2,2}$. The randomness of Alice includes $x_{1,1}, y_{1,1}, x_{2,1}, y_{2,1}$ and its correlated randomness R_1^g for the protocol Π_g . The randomness of Bob includes $x_{1,2}, y_{1,2}, x_{2,2}, y_{2,2}$ and its correlated randomness R_2^g for the protocol Π_g .

Protocol $\Pi_f(x, y)$:

1. Run protocol Π_g on input $(x, x_{1,1}, y_{1,1}, x_{2,1}, y_{2,1})$ for Alice and $(y, x_{1,2}, y_{1,2}, x_{2,2}, y_{2,2})$ for Bob and using correlated randomness (R_1^g, R_2^g) .
2. Let (x^*, y^*) be the output of Π_g . Output $f(x^*, y^*)$.

^aE.g., we can use Beaver's protocol that is based on random OT correlations [7]. The communication complexity and randomness complexity of this protocol are linear in the circuit size for g which is $O(n)$.

Figure 10: A WP 2-party protocol for any function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, in the correlated randomness model, with $O(n)$ communication and correlated randomness.

Due to the similarity with the DRE construction, we only sketch the proof of this construction.

Correctness holds because (x^*, y^*) is either (x, y) itself, or another input with the same f -value.

We argue the security with respect to Alice (the case of Bob is similar). Computing g itself reveals nothing beyond the output since Π_g is a secure protocol. We argue that the output gives WP. Recall that we denote the output of g by (x^*, y^*) . If $x \neq x^*$, then Alice knows that the input was (x_1, y_1) but she knows nothing about y_1 as this value was shared with Bob, so any y' such that $f(x, y') = f(x, y)$ is possible. If $x = x^*$, then again any such input y' is possible: it might be that (x^*, y^*) is the actual input (and comparison failed), but it can also be that $x_2 = x_1 = x$ and $y^* \neq y_1$ is just a random value where $f(x, y^*) = f(x, y)$ (since all values are shared these cases cannot be distinguished).

The complexity of our WP protocol (in the correlated randomness model) consists of running a standard

2PC protocol Π_g , followed by local computations by each party. We note that g is a simple function that has a simple circuit – comparing x to x_1 , which is the sum of 2 shares, and similarly comparing y and y_1 , and returning either (x, y) or (x_2, y_2) depending on the result. Each comparison is just the AND of n XOR gates, and each reconstruction of value from its shares is also n XOR gates. Therefore, g has a circuit of $O(n)$ size (and also low depth) and can be computed with $O(n)$ communication complexity using $O(n)$ OT correlations.

Thus, we obtain the following:

Theorem 5.8. *Let $f : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}$ be a function. Then, there is a $2^{-O(n)}$ -WP secure two-party protocol for f , in the correlated-randomness model with semi-honest parties, where both the communication complexity and the correlated randomness size are $O(n)$.*

Remark: Similar protocols can be constructed for general k -private, k -party MPC with correlated randomness, by using k -out-of- k secret sharing of the values. In this case, the input $\mathbf{x} = (x^1, \dots, x^k)$ is of total length $k \cdot n$, and sharing two such inputs $\mathbf{x}_1$ and $\mathbf{x}_2$ among the k parties will lead to $O(k^2 n)$ communication and correlated-random complexity.

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A A Lower Bound on the Number of OTs in WP 2PC

In this appendix, we show that the lower bound of [12] on the number of OTs in perfectly secure 2-party computation protocols applies also to WP protocols.

Theorem A.1. *Let $\text{EQ} : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}$ be the equality function. In any WP 2PC protocol for EQ in the OT hybrid model, the number of OT invocations is $\Omega(n)$.*

Proof. Let Π be a WP 2PC protocol for EQ in the OT hybrid model with k invocations of OT. As OT is symmetric [46], we can assume, without changing the number of OTs, that Alice gets all outputs of the OTs. We use Π to construct a deterministic non-private protocol Π' for EQ in which Bob sends one message of

length $2k + O(1)$ to Alice and Alice always outputs the correct output (the communication complexity of Π is independent of the communication complexity of Π' and only depends on the number of OT invocations in Π). By [47], the communication complexity of Π' is at least n , thus, $k \geq n/2 - O(1)$.

We next describe Π' .

- Alice's input: $x \in \{0, 1\}^n$, Bob's input: $y \in \{0, 1\}^n$.
- If $y = 0^n$ Bob sends 0 to Alice, and Alice outputs 1 iff $x = 0^n$.
- Fix a communication C that has positive probability when Alice holds 0^n and Bob holds 1^n .
- Bob finds the lexicographic first random string r_B such that Bob with input y and random string r_B sends messages as in C provided that Alice sends messages as in C (* we will show that such r_B always exists *).
- Bob simulates the execution of protocol Π with input y , random input r_B , and messages from Alice as in C . For each OT invocation in this invocation, Bob sends to Alice his two bits input to the OT.
- Alice find the the lexicographic first random string r_A such that Alice with input x and random string r_A sends its messages as in C provided that Bob sends messages as in C and the outputs of the OT invocations are computed using the bits that Bob sent (* as Alice has Bob's two inputs to each OT, she can compute the output of each OT and check if r_A is consistent *). If no such r_A exist or Alice outputs 1 in this execution, then Alice outputs 1 (i.e., $x = y$), otherwise she outputs 0.

We now prove the correctness of the protocol, only considering the interesting case in which $y \neq 0^n$. As $y \neq 0^n$ and $\text{EQ}(0^n, 1^n) = \text{EQ}(0^n, y) = 0$, the weak privacy of Π (as implied by the view of Alice) implies that C has positive probability given $0^n, y$; in particular, there exists a random string r_B such that Bob with input y and random string r_B sends messages as in C provided that Alice sends messages as in C .

Next, assume that $x \neq y$. As $\text{EQ}(0^n, y) = \text{EQ}(x, y) = 0$ and C has positive probability when Alice holds 0^n , Bob holds y and the random string r_B , the weak privacy of Π (as implied by the view of Bob) implies that C and r_B have positive probability given x, y ; in particular, there exists a random string r_A such that Alice with input x and random string r_A sends messages as in C provided that Bob sends messages as in C and the inputs to the OT invocations are as sent by Bob. Furthermore, by the correctness of Π , for every r_A, r_B, C that are consistent with x, y , the output of Π is 0, i.e., the output of Π' is correct when $x \neq y$.

If $x = y$, then either r_A as required by the protocol does not exist or, by the correctness of Π , for every r_A, r_B, C that are consistent with x, y , the output of Π is 1, i.e., the output of Π' is correct when $x = y$. $\square$